Business Dynamics Statistics for Single-Unit Firms

by

Richard Beem
U.S. Census Bureau

Christopher Goetz
U.S. Census Bureau

Martha Stinson
U.S. Census Bureau

Sean Wang
U.S. Census Bureau

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Abstract

The Business Dynamics Statistics of Single Unit Firms (BDS-SU) is an experimental data product that provides information on employment and payroll dynamics for each quarter of the year at businesses that operate in one physical location. This paper describes the creation of the data tables and the value they add to the existing Business Dynamics Statistics (BDS) product. We then present some analysis of the published statistics to provide context for the numbers and demonstrate how they can be used to understand both national and local business conditions, with a particular focus on 2020 and the recession induced by the COVID-19 pandemic. We next examine how firms fared in this recession compared to the Great Recession that began in the fourth quarter of 2007. We also consider the heterogenous impact of the pandemic on various industries and areas of the country, showing which types of businesses in which locations were particularly hard hit. We examine business exit rates in some detail and consider why different metro areas experienced the pandemic in different ways. We also consider entry rates and look for evidence of a surge in new businesses as seen in other data sources. We finish by providing a preview of on-going research to match the BDS to worker demographics and show statistics on the relationship between the characteristics of the firm's workers and outcomes such as firm exit and net job creation.

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1 Introduction

The Business Dynamics Statistics of Single Unit Firms (BDS-SU) is an experimental data product that provides information on employment and payroll dynamics for each quarter of the year at businesses that operate in one physical location. This paper describes the creation of the data tables and the value they add to the main Business Dynamics Statistics (BDS) product. We then present some analysis of the published statistics to provide context for the numbers and demonstrate how they can be used to understand both national and local business conditions.

The main distinguishing feature of the BDS-SU product is its quarterly frequency. Measuring business dynamics at different points throughout the year became especially important given the impact of the COVID-19 pandemic on employers. The main (core) BDS product, released in September 2022, provides annual statistics on employment changes from March year *t-1* to March year *t*, including 2019-2020 changes where only minor impacts of the pandemic on employment were evident. BDS-SU supplements the main BDS by capturing changes in quarters 2, 3, and 4 for the subset of firms where administrative tax data allow the measurement of later quarters. This timing enables the measurement of the impact of the pandemic on small businesses and complements the Census Bureau's Small Business Pulse Survey and Business Trends and Outlook Survey (Buffington, Chapman, Dinlersoz, Foster, and Haltiwanger, 2021; Buffington, Fields, and Foster, 2021).

Beyond quantifying the impact of the pandemic, these quarterly single-unit data have numerous advantages. First, because they operate in a single place, the firm, as defined by ownership, is definitionally equivalent to the establishment, as defined by location. This simplifies the reporting of business characteristics since age and size are, by definition, the same for the firm as for the establishment. Second, because the quarterly data allow us to identify the first and last quarters a business had employees, we can date business entry and exit more precisely during the year. Third, the quarterly data capture any large temporary disruptions to the economy that happen in a single year, such as a natural disaster or other transitory shock, which would be missed in the traditional BDS annual March-over-March job change estimates.[1]. Fourth, the quarterly data include firm-level payroll, allowing us to calculate payroll creation and destruction in an analogous manner to the employment statistics. Finally, it is relatively straight-forward to match administrative data on workers to single-unit firms and describe the demographics of the workers.

The BDS-SU is an experimental product of the U.S. Census Bureau and was developed by the Center for Economic Studies (CES). It is part of an on-going research agenda to expand the main BDS and is the third experimental product released since December 2021, following BDS-Goods Traders (Kamal and Ouyang, 2020; Handley, Kamal, and Ouyang, 2021) and BDS-High Tech (Goldschlag and Miranda, 2020). The published BDS-SU statistics were tabulated from the Longitudinal Business Database (LBD), an internal Census data product that tracks establishments over time (Jarmin and Miranda, 2002; Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson, and White, 2021). The LBD utilizes IRS quarterly tax filings at the Employer Identification Number (EIN) level to collect employment information for the pay periods including March

[1] For example, the 2021 main BDS which will be released in 2023 will likely miss much of the short-run impact of the pandemic because the economic rebound began so quickly

12th (Q1), June 12th (Q2), September 12th (Q3), and December 12th (Q4). We then calculate year-over-year employment changes between the same point in year *t* and year *t-1* (i.e., June 12th year *t-1* to June 12th year *t*, etc.). Based on the availability of quarterly employment data, the BDS-SU covers the time period 2007 quarter 1 through 2020 quarter 4 and is expected to be updated annually. The main reason to exclude multi-unit firms, i.e., firms that operate in multiple locations, is that this type of firm reports payroll and employment to the IRS at an aggregate (EIN) level so data are not available for individual establishments. The Census Bureau surveys a sample of multi-unit firms every year and conducts a census of most firms every five years to obtain quarter 1 data on the allocation of payroll and employment to individual establishments within the firm. These survey and census data are used to model the division of employment and payroll across locations for all multi-unit firms. However data are generally not collected for subsequent quarters, making allocation of EIN-level tax reports across different establishments infeasible for quarters 2 through 4.[2]

BDS-SU includes 44 tables, each with two versions. One version contains nominal payroll and the other contains real payroll defined in 2012 dollars. Tables are stratified by year, quarter, and firm characteristics, including geography (national, state, MSA, and county), industry (sector, 3-digit, and 4-digit NAICS), firm size, and firm age. Following the methodology and terminology of the main BDS, we label employment and payroll growth as job and pay creation and employment and payroll decline as job and pay destruction. In addition to employment and payroll dynamics, BDS-SU provides counts of operating firms in each quarter, split into categories of entrants, exits, and continuers in relation to the same quarter in the prior year. The combination of firm, employment, and payroll dynamics allows the measurement of the relative contribution to net job and pay gains of entering, exiting, and continuing firms, as well as the calculation of pay per employee for the different groups of firms. For a complete description of main BDS methodology see www.census.gov/programs-surveys/bds.html. A discussion of methodological issues specific to single-unit firms and a full list of BDS-SU tables available to download can be found at www.census.gov/programs-surveys/ces/data/public-use-data/experimental-bds/bds-single-unit.html.[3]

Given the large volume of statistics included, the goal of this paper is to provide a summary of the main trends observed in the data, with a special emphasis on the later quarters of 2020. We discuss how BDS-SU compares to the main BDS and how the 2020 quarter 2 recession compared to the recession that began in 2007 quarter 4. We also consider the heterogenous impact of the pandemic on different regions of the country and different industries, showing which types of businesses in which locations were particularly hard hit. We examine business exit rates in some detail and consider why cities experienced the pandemic in different ways. We also consider entry rates and look for evidence of a surge in new businesses as seen in other data sources such as the Business Formation Statistics (BFS) (Asturias, Dinlersoz, Haltiwanger, and Hutchinson, 2020; Bayard, Dinlersoz, Dunne, Haltiwanger, Miranda, and Stevens, 2018; Dinlersoz, Dunne, Haltiwanger, and Penciakova, 2021). We finish by providing a preview of on-going research to

[2]The Census Bureau is actively investigating ways to collect more quarterly data from firms by redesigning some traditional surveys. As these efforts are implemented, the LBD and BDS will make use of any new data to expand the published tables.

[3]This web site covers topics such as the precise definition of the single-unit universe, how we handle transitions from multi- to single-unit status, definitions for payroll creation and destruction, how we calculate quarterly firm age, and how we create real payroll.

match the BDS to worker demographics and show statistics on the relationship between firm outcomes, such as net job creation and exit, and the characteristics of the firm's workers.

Our findings corroborate the information reported in other sources such as the Census Bureau's Quarterly Workforce Indicators (QWI) and the Bureau of Labor Statistics's Business Employment Dynamics (BED). In 2020 quarter 2 there was an unprecedented drop in employment and rise in business exits. The fall in economic activity was much steeper than in the 2007-2009 recession, especially with respect to business exits. However, the BDS-SU tables document significant heterogeneity in the rates of decline. The largest drops happened in big cities and in-person service sectors. Yet even across cities, exit rates ranged from 24.2% to 8.8%, with differences related more to city-specific rates of business closure than to local exposure to hard-hit sectors such as hotels and restaurants. In spite of the severe economic contraction, recovery began quickly, with approximately a third of closed businesses re-opening by the end of 2020. New businesses also opened, although more commonly in states and sectors where not as many jobs were lost in 2020 quarter 2. These new businesses were not just self-employment ventures but also included firms who hired employees. Consequently, the number of firms operating for the first time reached its highest level since 2007 in quarter 4 of 2020. While more jobs were still being destroyed than created on net by the end of 2020, payroll growth had become positive, returning to pre-pandemic levels. Pay per employee showed evidence of early signs of inflation as it rose by over a thousand dollars in the latter two quarters of 2020. Finally, some types of workers were harder hit by the economic contraction than others. In particular, young, female, foreign-born, and non-white or Hispanic workers were especially likely to be working at firms that experienced net job losses or establishment closures in 2020.

2 Exploring Features and Trends in the Employment Dynamics of Single-Unit Firms

The BDS-SU data series allows us to explore the features of single-unit firms and trends in their employment dynamics since 2007. In this section we will begin by reporting some of the main statistics in the publicly available BDS-SU tables and comparing them to their counterparts in the main BDS tables. We will also look at the composition of the single-unit universe to describe what share of establishments and employment belong to various categories based on firm characteristics. This will be contrasted with the corresponding shares for the BDS universe as a whole to gain a sense of how representative the single-unit population is of the overall economy. Next, we will depict several trends in employment dynamics over the 2007-2020 period to reveal what sort of behavior and patterns the single-unit population has exhibited in relation to the existing BDS time-series. We will then use these plots to describe the differences between the recent COVID-19 recession of 2020 and the Great Recession of 2007-2009. Finally, we will dig deeper into the mechanics of the 2020 recession and explore how job creation and destruction differed according to factors such as firm size, geography, and industry.

2.1 Comparing Key Statistics in BDS-SU to the Core BDS

We first report several of the key economy-wide statistics for the single-unit universe in the first quarter of 2020. Table 1 reports the BDS measures for this period. Column 1 shows the value of each statistic in the BDS-SU, Column 2 shows the corresponding value from the core BDS in the same period, and Column 3 shows the ratio or difference between the two. Looking at Column 3 for the variable *emp*, we see that single-units represent 40% of the entire BDS universe in terms of employment. However, according to *estabs*, we see that single-units represent 72% of all the establishments in the BDS. This reflects the fact that establishments belonging to a single-unit firm employ fewer people on average than those owned by multi-unit enterprises. Yet despite representing less than half of overall employment, single-units are responsible for a disproportionately high share of the employment changes. As seen in the table, single-units account for 48% of job creation and 47% of job destruction compared to the overall BDS universe, meaning that single-units experience more employee turnover than multi-unit establishments do. This feature extends to establishment openings and closings as well. Single-units account for 85% of all establishment births and 84% of all establishment deaths, both significantly higher than single-units' 72% population share. Turning to the rate statistics in the lower panel of Table 1, we see that nearly all the rate of change measures are 1-2 percentage points higher for single-units than for the overall BDS. On net, the population of single-units accounted for the majority of the overall job growth rate from quarter 1 of 2019 to quarter 1 of 2020, with a net job creation rate of 0.6% comprising 61% of the 0.98% overall rate.

2.2 The Composition of the Single-Unit Universe

Clearly, the single-unit universe differs in some significant ways from the overall set of U.S. businesses. These firms are smaller and more dynamic, experiencing more growth and entry, yet also more volatile, experiencing more decline and exit. Other compositional differences further contribute to differences between the main BDS and BDS-SU tables. To see exactly how the composition of the single-unit population compares to the BDS universe as a whole, Table 2 shows what share of single-unit firms belong to various categories, based on size, age, geography, and industry. Columns 1 and 2 report the shares based on the number of establishments, for the BDS-SU and main BDS respectively. Similarly, Columns 3 and 4 show the respective shares weighted by the employment at those same establishments.

2.2.1 Age and Size

The top-left panel of Table 2 shows the shares of establishments belonging to each age category, and we see that single-units have a similar age distribution compared to the overall population of BDS establishments, never differing by more than a percentage point or two. However, when weighted by employment, as seen in columns 3 and 4, there is a more marked difference in the share of employment at establishments that are 11+ years old, implying that more mature multi-unit establishments employ more workers than their single-unit counterparts. The size categories in the second panel of the table are where we see the largest disparities between

single-units and the entire BDS universe. First, note that the vast majority of single-unit firms have fewer than 10 employees. While the 78% share of establishments with 1-9 employees is not that much higher than the 71% share in the BDS universe overall, the difference is more pronounced in terms of employment. While only 13% of workers overall are employed at establishments with 1-9 people, nearly one quarter of workers at single-unit firms work at this category of establishments. In the next smallest category, a full half of employees at single-unit firms work at establishments with 10-99 employees, compared to only 39% of employees overall. This relationship reverses in the larger two categories. For example, while 22% of workers overall are employed at establishments with 500+ employees, this is only true for 9% of those employed by single-unit firms.

2.2.2 Geography and Industry

Turning to the geographical location of establishments, the third panel of Table 2 shows that like the overall BDS universe, approximately 95% single-unit establishments and employment are located in metro areas, or Consolidated Business Statistical Areas (CBSAs) as defined by the Office of Management and Budget and the Census Bureau.[4] Similarly, looking at the bottom panel, we do not see large differences in the industry distribution between single-units and the overall BDS population, as categorized by 2-digit NAICS sectors. Single-unit firms are disproportionately classified in the Construction sector, at 12% of establishments and 9% of employment, versus 10% and 5% in the overall BDS respectively. In turn, single-unit firms are underrepresented in the Retail Trade, Information, and Finance sectors. Overall, however, the single-unit population does not appear to be heavily concentrated or skewed towards certain sectors of the economy relative to multi-units.

2.3 Trends in the Dynamics of Single-Unit Firms

We now turn to exploring trends in the employment dynamics in the single-unit population over the time span of 2007-2020. By plotting the time-series of the single-unit firms alongside that of the main BDS, we can compare the behaviors of the full and partial populations, and gain some insights from the additional data that the BDS-SU provides.

2.3.1 Employment

To see how employment has evolved in the BDS-SU, Figure 1 shows the time-series of the variable *emp* from 2007-2020 for the single-unit population (depicted with the solid blue line), compared to the core BDS (depicted in the dashed red line). Panel (a) displays the BDS-SU series on a quarterly basis, where we see significant fluctuations and a pronounced seasonal pattern. In

[4]While the BDS definition of metro area includes all 917 CBSAs according to the 2013-based OMB definitions, the BDS-SU only reports statistics at the individual metro-level for the 381 larger metros that are defined as Metropolitan Statistical Areas. The remaining Micropolitan Statistical Areas are aggregated into one category.

contrast, the core BDS is a smoother plot because it is only reported for quarter 1 in each year and thus has no within-year or seasonal variation. A key benefit of the BDS-SU series is that it extends three quarters beyond the core BDS in 2020. While the core BDS series ends with employment exceeding its 2019 levels in the first quarter of 2020 (March 12), the BDS-SU series reveals how dramatically employment fell in quarter 2 during the COVID-19 recession. Even though single-unit employment recovered somewhat in quarters 3 and 4, it still stood nearly 5 million short of its peak of 55.7 million in late 2019. Panel (b) of Figure 1 shows the BDS-SU employment level on a yearly basis, graphed separately by quarter. Here we see that the main reason for the seasonal variation in Panel (a) is due to the fact that quarter 1 employment is systematically lower than the other three quarters of the year, by approximately 2 million employees at all times. In turn, quarters 2, 3, and 4 are all comparable within a given year, although quarter 3 has generally recorded the highest employment in most years. While all four quarters follow the same general patterns in most years, we do see some differences in the timing of certain employment fluctuations, particularly during recessions. For example, because the economy peaked in the fourth quarter of 2007 before the Great Recession, we see it is the Q4 line that drops the most dramatically between 2007 and 2008. Similarly, because the COVID-19 recession largely hit right after March 12 in 2020, the Q1 line reveals a slight increase in employment, while the other three quarters show precipitous declines.

2.3.2 Job Creation and Destruction

Next, we turn to the key measures of employment change available in the BDS-SU, namely job creation and job destruction. Recall that all changes for a given period are calculated as the 4-quarter (or yearly) differences from the same quarter in the prior year. Figure 2 shows the time-series for *job_creation*, with Panel (a) expressing the changes in terms of the level of jobs, and Panel (b) expressing them as rates. The solid line represents the BDS-SU population, while the dashed line represents the overall BDS universe. We see that while single-units generally create fewer than half the jobs in any given period, they do so at a higher rate than in the overall BDS, implying that single-units are more dynamic than multi-units in their employment gains. Other than the seasonal variation in the quarterly series, both time-series tend to track one another closely, with some notable exceptions. While job creation among single-units was still peaking in the fourth quarter of 2007, it had already started to decline in the regular BDS. Conversely, the core BDS shows a run-up in job creation during 2016-2017 that is not apparent in the single-unit time series. Figure 3 shows the equivalent time-series plots for the variable *job_destruction*, with Panel (a) shown in levels and Panel (b) in rates. Again, while accounting for less than half of job destruction overall, the rates in the BDS-SU are consistently higher than those in the core BDS by about a percentage point. However, the job destruction rate increased more dramatically for single-unit firms, peaking at 22% in 2009 compared to 17% in the BDS universe overall. The job destruction rate for single-units reached new heights in the second quarter of 2020 at approximately 24% during the COVID-19 recession, and despite declining somewhat in the third and fourth quarter it remained similar to the rates of destruction during the Great Recession. Other smaller discrepancies are also apparent: job destruction rose in the core BDS in 2017, while it remained fairly flat among single-unit firms. Conversely, job destruction rose somewhat among single-unit firms in 2019 while it continued to decline in the BDS as a whole. The BDS-SU product will aid future research studying these differences in the

employment dynamics between single- and multi-unit firms.

2.4 Comparing the Pandemic Experience with the Great Recession

The BDS-SU data also allow data users to compare the more-recent pandemic experience with the Great Recession since the time series begins in the first quarter of 2007. Figure 4 plots quarterly net job creation for single-unit firms through the fourth quarter of 2020.[5] Several features of this graph stand out. First, the magnitude of the decline in net job creation during the pandemic was far more severe than it was during the Great Recession. Indeed, at their worst, single-unit firms were destroying roughly 4.4 million jobs year-over-year on net per quarter in mid-2009, more than 30% fewer than the initial decline recorded in the second quarter of 2020. That the decline in economic activity was less protracted during the more recent pandemic experience is not surprising given that much of the face-to-face economy was swiftly stymied by stay-at-home orders and the rapid spread of the COVID-19 virus. To this point, it is clear that the decline in net job creation during the Great Recession was more gradual. While it took six quarters for net job creation to reach its quarterly trough in the second quarter of 2009, during the pandemic this metric began rebounding immediately after the initial decline in the spring of 2020.[6] As stay-at-home orders were lifted and local economies began re-opening, temporary furloughs were reversed, putting upward pressure on net job creation. This structural difference in the *cause* of the two most-recent recessions helps explain the patterns seen in the data.

As net job creation is simply the difference between job creation and destruction, we can also glean more insights into the contrasting natures of the two recessions by looking at these components separately. Referring again to Figure 2 from Section 2.3.2, we see that the Great Recession can largely be characterized as a collapse in job creation, particularly among single-unit firms. The job creation rate in the BDS-SU universe dropped from nearly 20% down to 13%, and the overall job creation rate in the BDS fell from about 15% to 11%. While Figure 3 shows that the job destruction rate also went up during the Great Recession, it only rose by 2 percentage points overall and 5 percentage points for single-units. Therefore, the negative changes in job creation were nearly double the magnitude of the changes in job destruction. While we do not yet have the core BDS data necessary to study the COVID-19 recession, it is clear from the BDS-SU time-series in Figure 3 that the most recent contraction was primarily driven by job destruction. In Panel (b) we see that the job destruction rate for the BDS-SU population rose by 10 percentage points, from 14% to 24%, all during 2020 quarter 2. Both the magnitude and precipitous rise of the job destruction rate exceeded those experienced during the Great Recession. In turn, the job creation rate, as seen in Panel (b) of Figure 2, only fell by about 3.5 percentage points for single-units. Not only is this far smaller than the contemporaneous rise in job destruction, but it is also only about half as large as the 7% decline in the job creation rate experienced by single-units during the Great Recession. Therefore, the main explanations of the

[5] Although the frequency of the data is quarterly, the job creation and job destruction metrics used to calculate the net measure look back four-quarters to avoid confounding seasonal patterns.

[6] This initial release of BDS-SU data does not contain data from 2021 so we do not know the exact path of net job creation beyond 2020. However, total nonfarm employment began returning to its pre-recession level at a noticeably faster pace at the start of 2021. See https://fred.stlouisfed.org/graph/?g=WGkM. Should this feed into the single-unit universe in 2021, net job creation will become increasingly less negative.

COVID-19 recession will be more heavily centered on job destruction, in contrast to the Great Recession.

2.5 Heterogeneity: Firm Size, Geography, and Industry

Next, we dive deeper into our metrics of job creation and destruction. By how much did continuing single-unit firms contribute to total job creation and job destruction in 2020? What about new single-unit firms (entrants) and those that ceased to operate (exits)? Were these composition effects constant through the early months of the pandemic or did we experience a structural change of sorts? To answer these questions, we turn to Figure 5, which plots the share of job creation attributable to continuing firms and entrants as well as the share of job destruction attributable to continuing firms and exits. Importantly, these share decompositions are parsed out by firm size. For the purposes of this exercise, we define small single-unit firms as having fewer than 10 workers and large single-unit firms as having at least 10 workers.[7] Starting with the first two plots, continuing firms account for roughly 47% of total small-firm job creation. This share has remained relatively steady during the 2018-20 period. Interestingly, during the pandemic, small continuing firms recorded a marked increase in their contribution to small-firm job destruction. In the first quarter of 2020, the small continuing firms accounted for 45.1% of total small-firm job destruction. By the end of 2020, their contribution jumped to 50.1%. This pattern also appears for large single-unit continuers, but to an even larger extent. Indeed, single-unit continuing firms with at least 10 workers saw their contribution to total job destruction among large firms jump from 70.9% in the first quarter of 2020 to 81.7% by the end of 2020. What explains this shift in share of job destruction away from exiting firms and towards continuing firms? One possible explanation is the rollout of federal funds to assist businesses. Through the CARES Act, the Paycheck Protection Program (PPP) issued forgivable loans to businesses with fewer than 500 workers to help support the economy beginning in the spring of 2020. In order to be forgivable, businesses receiving these funds were required to maintain their payroll or hire back furloughed workers. If PPP loans prevented some single-unit firms from exiting, this would have helped shift the share of job destruction towards continuers. Another possible explanation is the nature of the slowdown in economic activity. At the beginning of the pandemic, many businesses were forced to temporarily halt operations due to the sudden onset of adverse conditions. But these closures were short-lived as businesses adapted and found ways to modify their operations and hence before the end of 2020 quarter 2, they had re-opened and reported employment and payroll on their tax filings.

As mentioned earlier, the pandemic brought about significant geographic heterogeneity in many metrics of business dynamism. We turn next to Figure 6, which presents a bivariate map at the county level showing the relationship between real payroll-per-worker in the second quarter of 2019 and the net job creation rate in the second quarter of 2020. This map helps us understand the pre-pandemic earnings characteristics of the places that were hardest hit by the pandemic. Was it mainly high- or low-pay-per-worker counties that bore the brunt of the surge in job destruction? For both real payroll-per-worker and the net job creation rate, counties are grouped by tercile. For example, counties with real payroll-per-worker in the bottom third

[7] Small single-units represent roughly 78% of all single-unit firms and nearly 25% of all single-unit employment.

appear in the left-most column of the legend, bound by the minimum and a threshold of $6,500 per quarter. Counties in the middle third report a real payroll-per-worker between $6,500 and $7,600 per quarter, while counties in the top tier have real payroll-per-worker between $7,600 and $18,200 per quarter. The number within each bin of the legend represents the percentage of all U.S. counties that fall in the category. Counties shaded in dark gold can be classified as having the highest income (from an earnings-per-worker perspective) and the most significant decline in their net job creation rate. These counties are clustered throughout the northeast corridor, which stretches from Boston to Philadelphia, and along the west coast. In contrast, much of the Midwest and Southeast are blanketed in shades of blue, suggesting that at least in the early months of the pandemic, job destruction in both high and low income counties in these regions was not yet severe. It is also notable that conditional on having a net job creation rate in the bottom third (most-negative), there is somewhat of a bimodal distribution in real payroll-per-worker. Indeed, 12.1% of counties were low income with another 12.1% classified as high income, but only 9.1 % falling in between.

To be sure, not all sectors and industries were affected equally during the pandemic. Building on the evidence presented in Figure 6, we now explore what sectors reported the largest loss of economic activity in 2020. Figure 7 plots each two-digit NAICS sector's net job creation rate from the second quarter of 2020.[8] Compared to the average decline in the net job creation rate of -12.2%, two sectors posted particularly large declines. NAICS 71, Arts, Entertainment, and Recreation, reported the largest decline with a 40% reduction. Not far behind was NAICS 72, Accommodation and Food Services with a decline of nearly 35%. These sectors operate primarily face-to-face and were most affected by the stay-at-home orders that were instituted in the beginning of 2020. It is also notable that nearly every sector posted declines in their net job creation rate. So while some sectors were extraordinarily affected, nearly all sectors destroyed more jobs than they created.

Diving deeper into the industrial composition of job loss during the pandemic, we turn to Figure 8, which focuses in on essential industries as defined by Blau, Koebe, and Meyerhofer (2021).[9] To accomplish this, we utilize the `bds2020_su_vcn4_real.csv` file and classify 197 four-digit NAICS industries as "essential." We then weight each industry's net job creation rate by its average-employment share to account for the scale of each industry. This produces a measure of each industry's contribution to the topline net job creation rate of -10.3% for all essential industries. Figure 8 shows that NAICS 7225, Restaurants and Other Eating Places, was the single-largest contributor to the overall decline in the net job creation rate among essential industries. It is followed by NAICS 7221, Traveler Accommodation and NAICS 6224, Child Day Care Services. Compared to all non-essential industries, which posted a net job creation rate of -16.6%, essential industries fared slightly better. This is not surprising since essential industries were exempt from some restrictions on business activity. To that point, being declared "essential" seems to have had some effect on measures of business dynamism. In the three years that preceded 2020, the essential-industries net job creation rate consistently hovered around 1.2 percentage points above its non-essential counterpart. This small but persistent gap in the

[8] Note that this graph omits NAICS 55, Management of Companies and Enterprises, due to its extreme volatility not often associated with underlying economic conditions, and NAICS 92, Public Administration, which is not included in the scope of BDS-SU.

[9] The authors use the guidelines issued by Department of Homeland Security (DHS) in March 2020 to identify NAICS industries considered "essential."

net job creation rate was only magnified during the pandemic. It is also worth noting that if Restaurants and Other Eating Places were not considered essential, the difference in the net job creation rate between essential and non-essential industries would have been even larger than the recorded gap of 6.3 percentage points.[10]

As a final source of heterogeneity, we now focus on the high-tech industries.[11] It is initially unclear how high-tech sectors might have fared during the pandemic. On the one hand, the services industries considered High Tech may have had the opportunity to continue their operations remotely, avoiding a major disruption in both employment and economic activity. On the other hand, the high-tech manufacturing industries were likely subject to slowdowns in production as the virus spread and workers became ill. To better understand how the high-tech sector fared in comparison to the non-high-tech sector, we turn to Figure 9. The first graph plots the net job creation rate for both the high-tech (green) and non-high-tech (blue) sectors. Two important features emerge. First, high-tech net job creation is persistently higher than that of the non-high-tech sector, suggesting that it is a valuable driver of employment growth. Second, while the non-high-tech sector recorded a dramatic decline in its net job creation rate, growth in the high-tech sector slowed rather than turning negative. And in the fourth quarter of 2020, its pace of job creation on net trended toward its pre-pandemic average. Not only does the high-tech sector create jobs at a faster rate than its non-high-tech counterpart, it also records stronger growth in inflation-adjusted pay. Indeed, single-unit high-tech firms recorded significantly positive real net pay creation rates even during the pandemic, finishing 2020 with a pay growth rate that exceeded the pre-pandemic average.

3 Pay Statistics in the BDS for Single Unit Firms

In addition to employment statistics, BDS-SU also provides quarterly statistics on firm-level payroll, which are not currently available in the main BDS series.[12] With these microdata we construct statistics on payroll dynamics in an analogous way to how the employment dynamics are calculated. Specifically, we measure the differences in payroll at the same firm between the current period and the same quarter in the previous year, labeling positive changes as payroll creation and negative changes as payroll destruction. Payroll creation can further be broken down between entrants and continuing firms, and pay destruction can similarly be divided between firm exits and continuing firms. Net payroll creation is defined as the difference between pay creation and pay destruction at the aggregate level. All these changes can also be expressed

[10] Indeed, if Restaurants and Other Eating Places were considered non-essential, then the net job creation rate for essential and non-essential industries would have measured -6.5% and -20.5%, respectively.

[11] For a discussion on the business dynamics of high-tech industries, see the following paper: Goldschlag and Miranda (2020) and also refer to the BDS-High-Tech tables, available here: https://www.census.gov/programs-surveys/ces/data/public-use-data/experimental-bds/bds-high-tech/data.html.

[12] Like the employment data, the IRS payroll measure is recorded on the aggregate level for multi-unit firms and is not easy to allocate across the different establishments. The BDS-SU process applies smoothing techniques to a small number of firm payroll records to correct for missing and erroneous data. Noise is also applied to the payroll data in a similar fashion to how it is done for the employment numbers in both the main BDS and BDS-SU. For more details on noise and general smoothing techniques see Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson, and White (2021).

in rates, where the denominator is based on the average of total payroll in the two corresponding time periods. Finally, the BDS-SU tables calculate payroll per employee, denoted as *pay_emp*, to provide a measure of average wages. Note, however, that this concept is complicated by the fact that while the employment count is a point-in-time measure based on a specific date within the quarter, payroll is the cumulative total of wages paid through the full quarter and may include wages for workers who are not counted at the reference date. Therefore, payroll per worker may be an overestimate of average wages.

Because payroll naturally rises over time, and since the BDS-SU time-series spans over a decade, we provide separate sets of tables based on nominal payroll and inflation-adjusted payroll respectively. Both sets contain identical employment-related statistics but differ in whether the pay-related measures are based on contemporaneous nominal dollars directly from the input data, or on an inflation-adjusted basis expressed in 2012 dollars. To apply the inflation adjustment, we use the GDP-based deflator constructed by the Bureau of Economic Analysis (Federal Reserve Bank of St. Louis , retrieved from FRED). All tables and payroll statistics with the suffix "_real" in their name are inflation-adjusted, and those without the suffix are in nominal terms.

Table 3 reports various payroll-related statistics from the first quarter of 2020. Total payroll in the single-units universe sums up to approximately 573 billion nominal dollars. While the main BDS does not report payroll, note for reference that the County Business Patterns program at the Census Bureau (which covers a similar scope of businesses as the BDS) reports total payroll as 2 trillion dollars for the same period. Thus, while single-unit firms account for about 40% of total employment, they only represent approximately 30% of total payroll, implying that the single-unit firms pay lower wages per worker than multi-units.

Other rows of the table show the payroll dynamics based on the changes relative to the first quarter of 2019. While we do not have statistics from the main BDS for comparison, we do notice that the payroll creation rate is higher than the job creation rate for the same time period in Table 1, and the payroll destruction rate is lower than the job destruction rate in the same table. This suggests that this was a period of significant growth in overall pay and pay-per-worker, as seen by the nominal net pay creation rate of 5.2% (and 3.6% in real terms). Note that overall pay-per-worker (*pay_emp*) is around $11 thousand per quarter in nominal dollars and $10 thousand in 2012 terms, suggesting an approximate average annual wage of around $40 thousand.

To see how these payroll dynamics have varied over time, Figure 10 shows the time-series of nominal and real net pay creation from 2007-2020. Panel (a) reports the net payroll changes in dollar-level terms, and Panel (b) reports them in growth rates (positive or negative). We observe that net payroll changes exhibit significant volatility in relation to net job creation. Also, note that because neither the nominal nor real pay data is seasonally adjusted, seasonal variations remain. However, the seasonal patterns are not as predictable as in the employment time-series. While Quarter 4 tends to be the highest in most years, this is not always the case. The fall in payroll during the COVID-19 recession is dramatic during the second quarter of 2020, but notice that net payroll creation recovers quickly and returns to positive territory by Quarter 4 – a feature not seen in net job creation.

Finally, we observe the trends in payroll-per-worker in Figure 11. The solid blue line represents overall pay-per-worker and we see that it has ranged from approximately $8 thousand

in nominal terms in 2007 to approximately $14 thousand by the end of the time-series. Notably there was a very sharp wage hike in the latter half of 2020, with pay-per-worker increasing by over a thousand dollars in just two quarters. This reflects the inflationary wage pressures that have been well-documented since the COVID-19 Recession. Note, however, that pay-per-employee follows a seasonal pattern that naturally tends to be highest in the fourth quarter of each year. Also, we see that pay-per-worker is largely driven by continuing firms, both because they represent the large majority of the firm population, and also because they pay higher wages per worker on average. Payroll-per-worker is consistently about 2-3 thousand dollars higher at continuing firms than at firms which are entering or exiting, with pay at exiting firms being a bit higher than pay at new firms.[13]

4 Firm Exit during the Pandemic

We now turn our focus to exit rates, particularly because this time series exhibited stark changes due to the COVID-19 pandemic. It is important to remember that exits, as defined in all BDS tables, are not necessarily permanent business closures or bankruptcies. An exit means that a business had positive payroll and employment the prior year but not the current year in the same quarter. A business can re-open after an exit, an event that we label as a reactivation. We begin by documenting national and sectoral trends in exit rates and then looking at differences across MSAs. We then decompose differences in MSA exit rates into the portion due to differences in exit rates and the portion due to differences in sector shares. Finally we end by discussing how to measure business reactivations and show results for the later quarters of 2020.

As shown in Figure 12, the national exit rate rose to 15.1% in quarter 2 of 2020, up 5.3 percentage points from 9.8% in quarter 2 of 2019, a rate which had been more or less constant since 2012. The onset of the pandemic caused a larger and more sudden shock to economic activity than the onset of the Great Recession. The exit rate then rose gradually between 2007 quarter 2 and 2009 quarter 2 from 11.3% to 13.0% before declining again in 2010.[14]

The national exit rate masks considerable heterogeneity. In Figure 13, we show the distribution of exit rates across NAICS sectors. The Arts-Entertainment-Recreation and Accommodations-Food Services sectors experienced the highest exit rates, both over 20%, or 5 percentage points above the national average. In comparison, the Professional-Scientific-Technical Services and Health Care had exit rates below the national average at 13.6% and 11.9% respectively. The

[13] Since payroll is $0 by definition when a firm exits, we assign firm exits the payroll figure from the corresponding quarter in the year prior to its exit.

[14] This rate corresponds to 774,710 exits of which 306,327 had been active in each of the three prior quarters and 192,240 of which remained inactive in 2020 quarter 4. In comparison, the Bureau of Labor Statistics Business Employment Dynamics (BED) time series reports 326,000 establishment deaths in 2020 quarter 2. These are establishments that operated in quarter 1 of 2020 and then had no employment or payroll in the following 4 quarters (through 2021 quarter 1). This includes establishments that belong to both multi-unit and single-unit firms so is expected to be larger than the most comparable BDS-SU number of 192,240 establishments that remained closed in 2020 quarter 4 for a third straight quarter. Note that the establishment death rate in the BED is not directly comparable to the BDS-SU exit rate both definitionally (BDS-SU is year-over-year and BED is quarter-to-quarter) and in terms of the universe of included firms (BDS-SU is just single-unit firms and BED is all firms).

Utilities sector recorded the lowest exit rate, unsurprisingly given the essential and stable nature of these businesses.

In addition to industry heterogeneity, there were also large differences across metropolitan statistical areas (MSAs). Figure 14 shows the distribution of exit rates across the 381 MSAs in the country, while Figure 15 plots the difference between each MSA exit rate and the national average. In Table 4 we list the 15 MSAs with the highest exit rates and show how much each exit rate changed relative to 2019 quarter 2 and how it compares to the national average. The New York City MSA experienced the highest rate of business closure in the country at 24.2%, a rise of 13.8 percentage points from the prior year and 9 percentage points above the national exit rate. This is unsurprising given the early arrival of COVID-19 in this city and the severity of the health impact. The next two cities on the list, Atlantic City, NJ and Lahaina (Maui), HI were likely severely impacted due to a drop in tourism. In general, the hardest hit MSAs were either on the East or West Coast. However, there were exceptions such as Detroit and Flint, MI which had exit rates 6 and 3 percentage points above the national average respectively. Even the MSAs with the lowest exit rates in 2020 quarter 2 saw increases from the prior year. Table 5 shows that the bottom 15 MSAs ranked by exit rate still saw their exit rates rise by 1-2 percentage points. Nevertheless, these smaller MSAs, largely in the Midwest and South, experienced exit rates at least 5 percentage points below the national average.

This spatial heterogeneity in exit rates leads one to ask what factors are driving the differences across MSAs. One hypothesis is that some MSAs were more exposed to the service industries which were particularly hard-hit during 2020 quarter 2. If sectors such as restaurants and tourism-related services were hard hit everywhere, then MSAs with a higher share of firms in these sectors would have higher exit rates. Another hypothesis is that differences in how localities experienced the pandemic drove differences in MSA exit rates. Metro areas had different COVID-19 infection and death rates, different responses by local governments, and different consumer behavior, by both local residents and out-of-area visitors. If these differences dominate, we would see varying exit rates in the same sector across MSAs. For example, the exit rate for restaurants in one MSA might be very high compared to restaurants in another MSA. While differences in industry shares and MSA-specific industry exit rates might both contribute to differences across MSAs, we will now investigate whether one cause is more dominant than the other.[15]

Figure 16 provides evidence about the first hypothesis and plots MSA exit rates against the share of MSA establishments in Accommodation and Food Services (sector 72). Here we do not see a strong relationship between share of MSA employment in the hospitality sector and the overall MSA exit rate. For example, New York City has a similar share of hotels and restaurants as Wausau, Wisconsin and yet New York City's exit rate is almost 25% whereas Wausau's is under 9%. In contrast in Figure 17, we compare the overall exit rate for each MSA to the sector 72 exit rate for the MSA. We see a much stronger relationship between the MSA exit rate in the hospitality sector and the overall MSA exit rate. Within the hospitality sector, the MSA exit rate varied widely, from 6% to 33% and this was highly correlated with the overall MSA exit rate. Firms belonging to sector 72 closed at a rate of 15% in Wausau, Wisconsin compared to 30% in New York City.

[15] We are only able to examine industry detail within state or MSA at the NAICS sector level. The BDS-SU does not contain any tables that cross 3 or 4-digit industry with geography.

To further quantify the effect of shares versus rates, we decompose the difference between each individual MSA exit rate and the average exit rate across all MSAs into a portion due to industry sector shares and a portion due to industry sector exit rates. The exit rate for a particular MSA is a weighted average of the exit rates for each sector which can be written as:

$$exitrate_m = \sum_j share_{m_j} \times rate_{m_j} \tag{1}$$

where

$$share_{m_j} = \frac{\text{total of all establishments in sector j in MSA m}}{\text{total of all establishments in MSA m}}$$
$$rate_{m_j} = \frac{\text{total of all exiting establishments in sector j in MSA m}}{\text{total of all establishments in sector j in MSA m}}$$

These MSA-specific exit rates and shares can be compared to average exit rates and average shares across all MSAs which are defined as:

$$\overline{share}_j = \frac{\text{total of all establishments in sector j across all MSAs}}{\text{total of all establishments across all MSAs}}$$
$$\overline{rate}_j = \frac{\text{total of all exiting establishments in sector j across all MSAs}}{\text{total of all establishments in sector j across all MSAs}}$$

We use these MSA-average shares and rates to define two hypothetical MSA-specific exit rates. The first replaces the MSA sector exit rates with the average sector exit rates as follows:

$$altr_exitrate_m = \sum_j share_{m_j} \times \overline{rate}_{m_j} \tag{2}$$

The second replaces the MSA sector shares with average sector shares as follows:

$$alts_exitrate_m = \sum_j \overline{share}_{m_j} \times rate_{m_j} \tag{3}$$

We can then compare all three MSA-specific exit rates to the average exit rate across all MSAs. First, we compare the actual MSA exit rate to the average and calculate the actual difference between these two measures. Next we compare the $altr_exitrate_m$ to the average and calculate an alternative difference. Finally we compare $alts_exitrate_m$ to the average and calculate a second alternative difference. How much and in what direction the MSA exit rate changes when we use average exit rates versus average shares tells what share of the full difference is due to each element. [16]

[16] We use the MSA by sector table to do these calculations. The published exit rates are calculated as the ratio of exiting

Figure 18 shows the results of this decomposition for the top 15 MSAs. The dark black bars show the total difference between the MSA exit rate and the average exit rate across all MSAs. The light hashed bars show how the difference would change if the MSA had exit rates equal to the overall MSA average in each sector. The height of this second bar is equal to the overall difference minus the portion due to differences in rates. The diagonal stripe bars show how the difference would change if the MSA had average shares of firms in each sector. The height of this bar is equal to the overall difference minus the portion due to differences in shares. In New York City, the MSA exit rate was almost 9 percentage points above the MSA average. However, if New York City sector exit rates had each been equal to the overall MSA average for those sectors, the New York City exit rate would have fallen to be almost equal with the MSA average exit rate. In contrast, if New York City sector shares had been equal to average MSA shares, the New York City exit rate would not have changed substantially. The relative importance of shares and rates differs across the MSAs shown in this figure but in general it appears that rates play a more important role than shares. When rates are set to the overall MSA average, the difference between the MSA exit rate and the average rate falls more than when MSA shares are set equal to the average. The same is true in Figure 19 for the MSAs with the 15 lowest exit rates. These MSAs had sector-specific exit rates well below the MSA average and if each of their sector exit rates had increased to the average rate for that sector, these MSA's overall exit rates would have risen to be almost the same as the average exit rate across all MSAs.

For some MSAs, the shares and rates push the MSA exit rate in opposite directions. For example if San Jose, CA had experienced average exit rates in each sector, its exit rate would be below the overall MSA average. In contrast, if its economy had been composed of average sector shares, its exit rate would have increased to be even further above the MSA average. In other words, San Jose's exit rate would have been even higher if it weren't for its below average shares in the most affected industry sectors. These shares partially offset the effect of the relatively high sector-specific exit rates.

Table 6 summarizes the directions in which rates and shares influence the exit rates of each MSA. Among the 64 metropolitan areas with exit rates above the overall MSA average, 35 of them had sector exit rates and sector shares that pushed their total exit rate higher. Among these 35, on average 76% of the difference between the MSA exit rate and the average exit rate

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firms divided by an average count of firms operating in the prior year and the current year. This denominator is referred to as *estabs_denom* in BDS documentation and is the basis for calculating shares. We re-create *estabs_denom* from the published MSA by sector table using the published exit rate, the count of exiting firms, and the definition of the exit rate. We sum *estabs_denom* across all included sectors in the MSA and then create sector shares as the ratio of *estabs_denom* for the sector and *estabs_denom* for the whole MSA. The MSA-level exit rate is calculated as the weighted average of the sector exit rates, where the weights are the sector shares. Because low exit rates in sectors 11, 21, 22, and 55 led to substantial amounts of cell suppression in the published table, we drop these sectors for each MSA for the purposes of this decomposition exercise. This leads to slightly different MSA-level exits rates than reported in Table 4. To be consistent with the earlier table, we report results for the same list of the top 15 MSAs, although in a slightly different order. To create a nation-wide average MSA exit rate, we first create an average MSA exit rate for each sector by calculating the share of each sector found in a particular MSA (ratio of MSA *estabs_denom* in the sector and total (sum of all MSAs) *estabs_denom* in the sector) and then use these sector shares to create the weighted average of MSA-sector-specific exit rates. These average exit rates by sector can be compared to MSA-specific exit rates. To create sector shares across all MSAs, we calculate the ratio of *estabs_denom* in a particular sector to total *estabs_denom* across all MSAs and sectors. These shares can then be compared to MSA-specific shares. The final nation-wide average MSA exit rate is the weighted average of the average sector-specific exit rates, using the across-MSA shares.

was due to higher sector-level exit rates. Another 15 of the 64 metropolitan areas had rates that pushed the exit rate up but sector shares that counteracted this to some extent, pushing exit rates down. A final 14 had shares that pushed the MSA exit rate up and rates that pulled it down. Thus for only 14 of the 64 MSAs, shares not rates were the dominating factor. For the 317 MSAs below the average exit rate, all but two were pushed below the MSA average by sector-specific rates. For 140 of them the shares pushed in the same direction. However, for these cases, 94% of the difference was accounted for by the exit rates. For the remaining 175, the shares tended to push the MSA rates up towards the average, counteracting the lower sector exit rates.

Many businesses close temporarily and then re-open, particularly during 2020. These within-year reactivations can be difficult to measure because of the year-over-year measurement rule in the BDS tables. A business that was closed during quarters 2 and 3 of 2020 would appear as an exit as long as it was active in quarter 2 and 3 of 2019. However, if it re-opened in quarter 4 of 2020, it would not appear as an entrant if it had also been active in quarter 4 of 2019. To capture these re-opening establishments, we rely on a count of the number of quarters of employment activity in the prior three quarters. Exiting firms in year *t* and quarter *n* who were active in all three prior quarters are closing for the first time in year t. For example in 2020 quarter 2, there were just over 306,000 exiting firms that had been active in quarters 3 and 4 of 2019 and quarter 1 of 2020 (see Figure 20). This number was three times as high as in quarter 2 of 2019 when only around 100,000 firms exited after three quarters of activity. Had all of these firms remained closed, we would expect to see a similar number of exiting firms in quarter 3 of 2020 with only 2 prior quarters of activity. These would be firms that were active in quarter 3 of 2019 but missed one quarter of activity between quarter 4 of 2019 and quarter 2 of 2020. However as shown in Figure 20, in 2020 quarter 3, the number of firms that remained closed after one quarter of inactivity was closer to 218,000, implying approximately 88,000 reactivations. [17] In quarter 4 of 2020, the number of firms with only one quarter of activity in the prior three quarters is approximately 190,000, implying a return to activity of an additional 30,000 firms. When the BDS-SU tables for 2021 are published, we will be able to measure the number of exiting firms in quarter 1 of 2021 with no quarters of activity between quarter 2 of 2020 and quarter 4 of 2020. This will give us the total number of businesses that closed at the beginning of the pandemic recession and had not re-opened by the same point one year later.

[17] The difference between quarter *n-1* exits with number of active quarters=3 and quarter *n* exits with number of active quarters=2 may slightly understate the number of reactivations. Some businesses with number of prior active quarters=2 may have been inactive in quarter *n-2*, then active in quarter *n-1*, and then returned to inactivity in quarter n. These cases will partially offset the fall in business exits by those inactive in the immediately preceding quarter (quarter *n-1*) returning to activity in quarter *n*. However the group of businesses that move in and out of activity multiple times within a year is likely to be small and hence number of quarters active=2 is likely to be mostly exits from quarter *n-1* that had been active until that quarter.

5 Firm Entry during the Pandemic

5.1 Comparing the BDS-SU to Business Formation Statistics data

As highlighted above, the quarterly frequency of the BDS-SU data allow for a more granular exploration of business dynamism during the pandemic. In particular, single-units reported massive declines in net job creation in the second, third, and fourth quarters of 2020 that rivaled and even exceeded the declines experienced during the Great Recession. Pairing the BDS-SU data with the Business Formation Statistics (BFS), we can explore the extent to which the hardest-hit states and sectors also reported a subsequent rebound in business application activity during the second half of 2020. Was there a widespread replacement of business activity in the form of new enterprises? Did states like New York and New Jersey see meaningful rebounds in business application activity? In the Southeast and Midwest, where COVID case counts remained initially low, did the magnitude of business starts look materially different from much of the pre-pandemic period? Turning to the hardest-hit sectors, did business activity in the arts and restaurant sectors recover with the startup of new enterprises? Was there a potential shift in the aggregate industry composition during the pandemic relative to the years prior? This section explores these important questions empirically.

These important questions would be difficult to answer if not for the efforts of Bayard, Dinlersoz, Dunne, Haltiwanger, Miranda, and Stevens (2018), who constructed the public-use BFS dataset using IRS administrative data on applications for Employer Identificaton Numbers (EINs). Businesses apply for EINs in anticipation of eventually filing payroll and revenue tax reports with the IRS and thus these applications are leading indicators of business entry. This research fueled subsequent efforts to better understand how business formation during the Great Recession compared to the most-recent pandemic-induced recession (Dinlersoz, Dunne, Haltiwanger, and Penciakova, 2021). More broadly, research utilizing the BFS shows that high-propensity business applications, or those that are likely to transition into an employer business, are a leading indicator of general economic activity (Asturias, Dinlersoz, Haltiwanger, and Hutchinson, 2020). And more recently, Decker and Haltiwanger (2022) show that the pandemic-induced changes to work and lifestyle patterns resulted in cross-industry restructuring.

To explore whether the hardest-hit states also experienced a replacement of lost business activity in the form of new enterprises, we perform the following data exercise. We first construct a state-level BFS dataset at the quarterly frequency by aggregating the publicly available monthly data.[18] These data are then merged with the nominal state-level BDS-SU file to form a balanced panel starting in 2007q1 and ending in 2020q4.[19] For the purposes of this exercise, we will focus on the job destruction rate from 2020q2 and the subsequent business application growth rate from 2020q3.[20] We define the business application growth rate in quarter q and year t as $rate_{qt} = ((apps_{qt}/apps_{qt-1}) - 1) \times 100$. This four-quarters-ago comparison ensures that our

[18] These data can be found here: https://www.census.gov/econ/bfs/csv/bfs_monthly.csv.

[19] The nominal state-level BDS-SU file is bds2020_su_st_nominal.csv.

[20] We also considered using the business application growth rate from 2020q4, but the 52nd and 53rd week of data in 2020 are lumped together and cannot be parsed out to compare with the prior year, which was only 52 weeks. Nevertheless, ignoring this issue and proceeding with this data exercise yields economically similar results.

measures of business application growth will be comparable to the pre-defined job destruction rate. Importantly, these rates will be standardized with a mean value of zero and a standard deviation equal to one. This exercise is repeated for sectors at the national level.

Figure 21 reports the state-level results of this exercise. Each of the three graphs plots the standardized job destruction rate from 2020q2 on the horizontal axis against the standardized business applications growth rate from 2020q3. Each bubble represents a specific state with its size determined by denom, the average of employment in year t and $t-1$. Thus, larger states are represented by larger bubbles. For states positioned to the right of the vertical zero line, their job destruction rate was higher than average. Similarly, states positioned above the horizontal zero line recorded higher-than-average growth in business application activity. The first graph plots the business application growth rate for *all* business applications, while the second looks specifically at high-propensity business applications, or those considered "likely employers." The third graph looks at an even finer subset of total business applications: those with planned wages, a specific type of high-propensity business applications. The latter two graphs are perhaps most important because business applications of the high-propensity type are likely to transition into the single-unit universe of existing firms.

Across all three graphs in Figure 21, it is clear that New York and New Jersey, which are positioned farthest to the right, reported the highest job destruction rates in 2020q2, measuring 31.7% and 30.7%, respectively. Interestingly, their subsequent growth in business application activity is not meaningfully different from the average across states. In fact, very few hard-hit states experienced a sizeable rebound in business application activity, with the exception of Michigan. Mississippi, Georgia, Louisiana, and Illinois, which fared quite well from a job-loss perspective, are the states that fueled the summer-2020 surge in new business application activity. Concentrated in the Southeast and Midwest, these states were not particularly hard hit with surging COVID case counts. This finding runs counter to the intuition that geographies recording the most loss should also record the highest growth in new business enterprises. At first glance, it appears that this "replacement" story does not hold. To the extent that there was sizeable migration during the pandemic, it could be the case that displaced workers moved across state lines and started new businesses there. Decker and Haltiwanger (2022) do find that within major cities, business applications, net establishment births, and excess job separations exhibit a "donut" pattern with more robust growth in the surrounding area. The finding here is similar, but on a more aggregated scale.

Turning next to the sector-level findings in Figure 22, similar results emerge. Starting with the first graph, it is clear that three sizable sectors stand out: retail trade (44-45), arts, entertainment, and recreation (71), and accommodation and food services (72). Not surprisingly, the BDS-SU data reveal the highest rates of job destruction among the latter two sectors. These in-person services fared poorly when local economies were effectively shut down and stay-at-home orders were instituted during the early months of the pandemic. Conversely, the job destruction rate in retail trade measured just shy of the average across all sectors. However, its subsequent growth in business application activity proved extraordinary. Indeed, the overall surge in total business applications during the summer of 2020 was predominantly fueled by retail trade, non-store retail in particular. Again, it appears that the hardest-hit sectors were not the main beneficiaries from heightened growth in business application activity in late 2020. It is notable that in the high-propensity graph the accommodation and food services sector finds itself firmly

situated in the first quadrant. The interpretation here being that although its job destruction rate was much higher than the average, it also recorded above-average growth in likely employer businesses. Pending their potential single-unit status, these new business ventures could be viewed as a replacement of lost business activity. It is worth noting that displaced workers from the two hardest-hit sectors could very well have started online retail businesses during the pandemic. It is unclear whether this phenomenon occurred more frequently than the migration story outlined above, but both offer potential explanations for the findings in the data.

The data exercise presented in this section, which pairs the BFS with the BDS-SU, offers a high-level glimpse at how the most-affected states and sectors initially began rebounding from historic levels of job destruction. Most interesting is the lack of correlation between severity of job loss and subsequent rebound in business application activity. As mentioned above, the migration of displaced workers might explain the state-level findings while the sector-level result might be better explained by displaced workers starting businesses in tangential sectors. There are questions that still remain, however. Future research in this area might examine whether the timing and magnitude of stimulus checks aided in the startup surge that still persists today. It could very well be the case that many of these startups turn into single-unit firms. Whether these new business ventures were devised to simply take advantage of a major market disruption should, in part, help determine their future prospects and longevity. The BDS-SU is well positioned to measure how many of these business applications turn into actual employer businesses, a topic we turn to now.

5.2 New employer entry: signs of business recovery

Large numbers of applications to start businesses do not necessarily translate into more businesses that hire employees. To examine the trends in the formation of new employer businesses, we turn to our firm age table to identify age 0 firms. The BDS-SU innovation of measuring firm age in quarters instead of years allows us to identify firms that were operating in their first quarter ever. Figure 23 plots the share of operating firms that are age 0 over time, separately by quarter. Quarter 1 always has the highest share of new employer firms, a fact that is consistent with other data sources but which likely reflects seasonal trends.[21] The share of firms that were age 0 in quarter 2 began at 2.5% in 2007 and fell to 1.9% in 2009. It rose slightly at the end of the 2010s but fell to 2.0% in quarter 2 of 2020, a decline of 17%. However in quarters 3 and 4 of 2020, the share of new firms rose relative to 2019. By quarter 4, the share of age 0 firms was 2.5%, higher than the fourth quarter in any other year in the time series and a rise of 8% relative to 2019 quarter 4.

While it is true that the total number of establishments operating in quarter 4 of 2020 declined relative to 2019, this was not the sole cause for the rise in the share of age 0 firms. The number of firms operating for the first time ever in quarter 4 rose from 124,428 in 2019 to 134,860 in 2020, the highest level in the 2007-2020 time series. Thus there was some uptick in new employer business formation towards the end of 2020, consistent with the rise in ap-

[21] BFS and the Business Employment Dynamics (BED) both see the highest share of new firms in quarter 1. This is possibly related to requirements for filing taxes in April each year and hence should be viewed as a seasonal pattern and not necessarily an indicator of economic activity.

plications for EINs observed in the BFS time series. It is, however, interesting to note that the employment created by these new firms comprised a similar share of total employment in both 2020 and 2018 (1%) and the level of employment was lower in 2020 than in 2018 (527,000 versus 534,433 jobs). Real payroll at start-ups made up a lower share of total payroll in 2020 than in 2018. The new businesses forming in 2020 were smaller in terms of their contribution to job and payroll creation.

Figure 24 shows how the distribution of age 0 firms changed between 2019 quarter 4 and 2020 quarter 4. Notably Retail Trade, Transportation-Warehousing, and Real Estate made up a higher share of age 0 firms in 2020 compared to 2019. Not surprisingly Arts-Entertainment and Accommodation-Food Services represented lower shares of age 0 firms. It is worth noting that these changes are not dramatic but rather slight shifts. In 2020 quarter 4, Construction and Professional-Scientific-Technical Services still made up 11.1% and 16.3% of age 0 firms respectively, down only slightly from 2019 quarter 4. Even Accommodation-Food Services remained at 8.8%, down from 10.1%. Still pandemic headwinds shifted employer business formation out of sectors that sold services to customers and into sectors related to selling products.

Finally in Figure 25, we show how the distribution of age 0 firms across states changed between 2019 quarter 4 and 2020 quarter 4. New York saw the biggest drop in its share of total start-ups in the country, falling from 7.4% to 6.6%, a change of .8 percentage points, or 10% of the 2019 level. California remained the state with the highest share overall, 16%, but fell .35 percentage points or 2%. Florida, Texas, Georgia, North Carolina, New Jersey, and Utah saw increases in their shares of age 0 firms. Utah in particular saw a 13% rise in its share, from 1.7% to 1.9%. Given that Utah represents only 1.2% of all single-unit firms, its start-up share is higher than would be expected. Florida is even more dramatic, representing 9.8% of start-ups but only 7.8% of single-unit firms. As with industry, the geographic shift in where new employer firms formed was slight but still consistent with trends seen elsewhere. New business formation shifted out of traditional locations in big cities in coastal states and into more southern, less densely populated states.

6 Worker exposure to business dynamics at single-unit firms

Given the systematic differences in business dynamism between different types of firms that we have documented in the BDS-SU data, a natural question is how these differences ultimately affect various types of workers. Researchers emphasize that labor markets are segmented in dimensions that are not captured in standard firm data (Altonji and Blank, 1999; Bayer, Ross, and Topa, 2008). A substantial portion of the dispersion in firm economic performance and in management practices also exist within industries or geographies (Syverson, 2011; Bloom, Brynjolfsson, Foster, Jarmin, Patnaik, Saporta-Eksten, and Van Reenen, 2019). As a result, existing firm- or worker-level data products are unlikely to capture the full heterogeneity in firm-level economic conditions that are experienced by different types of workers.

In this section, we describe the methods and results of a new, complementary research project that documents the firm-level correlations between business dynamism and workforce compo-

sition. We leverage the relatively simple organization of firms in the BDS-SU universe to create a new dataset linking nearly all BDS-SU establishments to the universe of their wage-earning employees through administrative individual tax data. This dataset connects data on firm-level dynamics from the LBD to data on individual-level characteristics from the Census Bureau and the Social Security Administration (SSA). We use our new dataset to calculate the workforce composition at each firm in our BDS-SU universe, and we describe the differences in average business dynamism between firms that tend to hire different types of workers. In addition to contributing valuable new insights about labor market segmentation, we hope this new worker-linking project will also serve as the foundation for improvements to the longitudinal linkages in the LBD as well as for a future BDS - Human Capital experimental product that reports statistics about workers employed at firms in various categories.

We highlight three key initial findings from the linked worker data. First, different types of workers are exposed to systematically different economic environments through their employers. The differences in average firm net job creation rates and establishment exit rates by firm workforce average age, female share, foreign-born share, or non-white or Hispanic share are comparable in magnitude to the differences between firms by sector or firm age. Second, overall job losses and establishment exits in 2020 were concentrated in firms that employed more young, female, foreign-born, and non-white or Hispanic workers. This stands in contrast to the distributional impact of business dynamics both during the Great Recession as well as during the 2010s. Third, these differences in the average exposure to net job creation or establishment exits are not driven by any single worker group or by differences in the firm industry or geography. Differences in the exposure to net job creation tend to become larger after controlling for the firm's industry and geography, which underscores the value of tracking worker demographics at the firm-level rather than at more aggregate levels.

6.1 Methodology

To describe the characteristics of workers at each firm, we construct a new research dataset that links the underlying LBD establishment-level microdata for BDS-SU firms to the universe of their wage-earning employees using administrative IRS W-2 records. These tax records are filed by businesses to report total earnings individually for each worker employed during the calendar year. Since the LBD is constructed from business tax filings that report aggregate employment and payroll each quarter (Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson, and White, 2021), the underlying frames of firms in the LBD data and the W-2 data are conceptually aligned, and it is straightforward to match workers to firms by EIN. Furthermore, since the BDS-SU universe focuses on the set of firms that contain a single establishment, these worker-firm links also correspond to worker-establishment links.

We define a job as a unique worker-LBDNUM record to leverage the longitudinal linkages made by the LBD to track reorganizations and EIN changes. When an establishment reorganizes and changes EIN, workers' earnings may be split across the old and new EIN in the W-2 records during a single calendar year. The LBD tracks these within-year EIN changes through the ein_rorg variable, and we use ein_rorg to aggregate worker earnings across these within-year reorganizations into an establishment-level total. Similarly, across-year reorganizations

may cause workers who remained at the same establishment to change EINs in the W-2 data. By mapping EINs each year to the longitudinally consistent LBDNUM identifiers, we can separate true job changes from spurious EIN changes.

The linked W-2-LBD data provide a full list of workers who were employed at any BDS-SU firm in each year. We add worker demographic characteristics to these linked data by merging in individual-level characteristics from the SSA NUMIDENT, the 2010 Decennial Census Edited File, and the LEHD Individual Characteristics File (ICF) using the worker's Protected Identification Key (PIK). The coverage of workers in each dataset varies due to the timing of data collection and the universe of individuals included. In addition, the availability of demographic fields varies by dataset. When multiple sources of data exist for a worker's demographic values, we take the SSA NUMIDENT information when available, and supplement missing observations first with the 2010 Decennial Census and then with the LEHD ICF.[22]

We consider four key, time-varying workforce characteristics: average worker age (in decades), the share of female workers, the share of workers born outside of the U.S., and the share of non-white or Hispanic workers. For each workforce characteristic, we calculate the firm-level average as the simple mean over all workers who received W-2 income from the firm in each year, including part-time and partial-year workers. We also construct a measure of large firms' earnings inequality in each year as the ratio of the 90th percentile and the 50th percentile W-2 earnings at the firm. To help mitigate concerns that variation in earnings inequality may be driven by small firms and partial-year workers, we define the measure based only on the earnings of "stayers" — workers who received W-2 earnings from the same establishment both in the prior year and in the subsequent year.[23] We further restrict the definition of earnings inequality to firms with at least 10 such stayers and where median stayer earnings are at least $5000 in 2012 dollars, and we winsorize the inequality measure at the 5th and 95th percentiles to downweigh the effect of outliers.[24]

The focus of our analysis is to describe how different types of workers may be differentially exposed to firm job creation or establishment exit. We implement this analysis by estimating two complementary regression specifications on a random 10% sample of our BDS-SU universe of firms. The first regression is the quarterly regression specification

$$y_{it} = \beta_\tau x_{it} + \gamma_\tau + \varepsilon_{it} \tag{4}$$

where y_{it} is either the year-over-year net job creation rate or an indicator for exit for establishment i in quarter t, and x_{it} is a single establishment workforce characteristic.[25] Our coefficients of interest are β_τ, which we allow to vary flexibly by quarter. We also include quarter fixed ef-

[22] We prioritize data in this sequence to maximize the coverage of administrative data and to ensure that the LEHD imputations are used only in the case that we are missing both SSA and Decennial Census data.

[23] This is similar to the definition of "full-quarter" workers in the Quarterly Workforce Indicators (QWI). Since our W-2 data are at the annual level, we cannot verify directly whether each "stayer" worked all four quarters in a given year. So, we instead use the term "stayer" to help clarify that our definition is not necessarily a strict refinement of the QWI definition.

[24] Defining "stayers" requires data from both the prior year and the subsequent year. Due to this data limitation, we do not calculate earnings inequality in the first and last years of our sample.

[25] As we discuss later in the subsection, the workforce characteristic is generally defined using data from two years prior to avoid mechanical biases. For entrants, we generally use data from the year after entry.

fects γ_τ to flexibly control for aggregate business cycle trends.[26] To further decompose the correlation between each workforce characteristic, we also pool effects over the entire 2007–2020 sample period and estimate the joint regression specification

$$y_{it} = \Theta X_{it} + \gamma_\tau + \alpha_j + \eta_g + \varepsilon_{it} \tag{5}$$

where y_{it} is again either the net job creation rate or an indicator for exit for establishment i in quarter t, and X_{it} is the full set of all workforce characteristics that we consider. Our coefficients of interest is the vector Θ, which is the average correlation for each worker characteristic over the sample period. We still include quarter fixed effects γ_τ to control for aggregate business cycle trends, and in some specifications, we also add 6-digit NAICS fixed effects α_j and state-by-MSA fixed effects η_g to flexibly control for the firm's industry and geography. We weight all regressions by average quarterly employment in the current and prior year (*bds_denom*) for consistency with the aggregate tables in the BDS and BDS-SU, and we cluster standard errors by industry to allow for arbitrary correlation in establishment unobservables over time and within industries.

We emphasize that our empirical models are descriptive, and the interpretation of their coefficients is the differential exposure of different workers to business dynamism rather than the causal effect of different types of workers on firm growth or exit.[27] The composition of workers at different firms may be determined by many factors that are correlated with the firm's economic performance. Research designs that do not control for these factors are unlikely to identify the underlying causal relationship.

Although our objective is to estimate the descriptive relationship between workforce characteristics and business dynamism, we do stagger the timing of business outcomes and workforce characteristics in our regressions to ensure that our results are not mechanically determined by business outcomes. For continuing and exiting firms, we compare their firm job growth rates between the current and previous year to their workforce characteristics from two years prior. We assign entering firms their workforce characteristics from the year after entry.[28]

6.2 Results

Figure 26 plots the average relationship between a firm's overall net job creation rate and each of its workforce characteristics over time. Each point on the line is the quarterly coefficient from estimating Equation 4 using the corresponding workforce characteristic. Older workers were less exposed to net job creation relative to younger workers over the 2010s. For most of the decade, firms with workforces that are 10 years older on average experienced 3 percentage

[26] Note that this specification is equivalent to estimating the simple bivariate regression $y_{it} = \beta_\tau x_{it} + \varepsilon_{it}$ separately for each quarter τ. Estimating the coefficients jointly allows us to cluster standard errors across quarters.

[27] This is similar to how outcomes in the BDS products reflect descriptive statistics of activity in each firm category (e.g., young firms or manufacturing firms) rather than the causal effects of those characteristics on firm growth.

[28] In the small number of cases where we do not observe workforce characteristics from the relevant year, we use workforce characteristics from the prior year. If data are also unavailable from the prior year or if the establishment is an entrant, then we use data from the contemporaneous year.

points (p.p.) lower year-over-year net job creation rates in each quarter. Female workers were also less exposed to net job creation relative to male workers over the same period. Firms with workforces that have a 50% higher share of women experienced up to 1.5 p.p. lower year-over-year net job creation rates in each quarter. On the other hand, foreign-born workers and non-white or Hispanic workers were generally more exposed to net job creation than U.S.-born workers and white non-Hispanic workers. For most of the decade, firms with workforces that have a 50% higher share of foreign-born workers or non-white or Hispanic workers had up to 2.5 p.p. higher net job creation rates in each quarter. Finally, firms with relatively higher earnings inequality had net job creation rates that were similar to firms with lower earnings inequality, so the overall difference during the 2010s was negligible.

We can benchmark the magnitude of differences in Figure 26 by comparing them to the net job creation rates in various BDS-SU tables. From the BDS-SU Sector table, the Information sector (one of the fastest-growing sectors) had an average net job creation rate of 4.6 p.p. between 2011 and 2019, while the Wholesale Trade sector (the slowest-growing sector) had an average net job creation rate of 1.2 p.p. over the same period. From the BDS-SU Firm Age table, firms that were created 4–20 quarters (1–5 years) ago had an average net job creation rate of -0.9 p.p. between 2011 and 2019, while firms that were created before 1976 had an average net job creation rate of -1.5 p.p. over the same period. Therefore, the range of differences in net job creation rates by firm workforce characteristics is at least comparable in magnitude to the range of differences by firm sector or firm age.

Figure 26 also highlights that the differences in worker exposure to business dynamics are magnified during recessions and that 2020 was an especially unusual year even when compared to the Great Recession. Relative net job creation rates for firms with more female workers, foreign-born workers, or non-white or Hispanic workers plummeted during 2020, whereas relative net job creation rates for firms with more older workers spiked. In other words, adverse firm employment conditions in 2020, including net headcount reductions or firm closures, were concentrated on younger workers, female workers, foreign-born workers, and non-white or Hispanic workers. The same patterns did not exist during the Great Recession. Relative net job creation rates were substantially higher in 2008 and 2009 for firms with more female workers and foreign-born workers, and close to 0 for firms with more older workers and non-white or Hispanic workers.

To better understand the differences in business dynamics, Figure 27 plots the average relationship between a firm's exit probability and each of its workforce characteristics over time.[29] Some salient features of the figure are consistent with Figure 26 — firms with more female workers were less likely to exit during the Great Recession and firms with more female workers, foreign-born workers, or non-white or Hispanic workers were more likely to exit in 2020. But the figure also shows key differences between net job creation and establishment exit. During the 2010s, older workers were exposed to fewer establishment closures despite being exposed to lower overall net job creation, while foreign-born workers and non-white workers were more exposed to both more establishment closures and higher net job creation. This suggests that foreign-born workers and non-white workers may face generally riskier economic environ-

[29]Note that although the regression specifications are identical between Figures 26 and 27, the outcome variables differ by sign. All else equal, establishments that are more likely to exit have lower net job creation and higher exit rates.

ments. Meanwhile, although high inequality firms have similar levels of overall net job creation as lower inequality firms, they are less likely to exit both during the Great Recession and the 2010s.

Our results so far have highlighted that different groups of workers are exposed to systematically different economic environments, both during the 2010s and especially during economic downturns. We next turn to the joint regression model specified in Equation 5 to show that these differences between firms by workforce characteristics are not driven by any single group of workers or by firm differences in industry or geography.

Table 7 pools all quarters in our sample and reports the overall joint relationship between a firm's net job creation rate and each of its workforce characteristics. Column 1 shows that over 2007–2020, firms with workforces that are 10 years older have 2 p.p. lower average year-over-year net job creation rates, while firms with a 50% higher share of foreign-born workers have 2 p.p. higher average year-over-year net job creation rates. Both differences are statistically significant at the 0.1% level. On the other hand, firms with a higher share of female workers do not differ meaningfully in net job creation rates, and firms with a 50% higher share of non-white or Hispanic workers have 0.6 p.p. lower year-over-year net job creation rates. Labor markets are segmented, so firms with different worker characteristics are also likely to be in different industries and regions. To understand whether industry and geography differences can explain the differences in net job creation rates in Column 1, we focus our comparison between firms in the same industry (by adding 6-digit NAICS fixed effects) in Column 2 or firms in the same industry and geography (by adding both 6-digit NAICS and state-by-MSA fixed effects) in Column 3. We find that differences between firms *increase* when we compare observationally similar firms — the relative net job creation rate for firms with a 50% higher share of female workers falls from 0 to -1.4 p.p., and the relative net job creation rate for firms with a 50% higher share of non-white workers falls from -0.6 p.p. to -1.1 p.p. These results imply that rather than explaining the raw differences in Figure 26, industry and geography differences for workers are masking the larger within-industry and within-geography differences in their exposure to firm net job creation. Finally, Column 4 additionally considers earnings inequality in the subsample of larger firms. We find that on average, firms with higher earnings inequality experience net job creation rates that are similar to other firms.[30]

Table 8 repeats the joint analysis for establishment exit and further emphasizes that establishment exit is a distinct outcome from net job creation. Comparing Column 1 to Columns 2 and 3 shows that unlike the case of net job creation rates, the average relationship between establishment exit rates and workforce average age reverses signs after controlling for industry. Firms with workforces that are 10 years older on average are 0.12 p.p. more likely to exit in a given year when we compare firms in the same 6-digit NAICS industry or the same state and MSA. We do not find statistically significant differences in establishment exit rates by the share of female workers, either with or without industry and geography controls. Our results by foreign-born share and by non-white or Hispanic share are similar to our net job creation results — non-white workers are more exposed to firm exit while foreign-born workers are less exposed to firm exit. Finally, Column 4 shows that firms with higher earnings inequality are less likely to exit in a given year even after controlling for other worker characteristics and the firm's industry

[30] We also report the coefficients for the other four worker characteristics for completeness, but the estimates are not directly comparable to the estimates in other columns since the sample and the set of regressors are both different.

and geography.

Finally, Table 9 combines the results in Tables 7 and 8 to show the relationship between workforce characteristics and the components of net job creation not attributed to establishment exit. By setting net job creation to 0 for exiting firms, this analysis focuses on the components of net job creation that are due to firm entry or the job growth at continuing firms. The regression specifications in the table are otherwise identical to Table 7, so the coefficients are directly comparable. Comparing Column 3 of Table 7 to Column 3 of Table 9, the estimated relationship between firm net job creation and the non-white or Hispanic worker share falls by more than half when excluding establishment exit. So, a majority of the within-industry and within-geography exposure of non-white workers to lower net job creation is due to the higher likelihood that their employers will exit. On the other hand, the coefficient on earnings inequality in Column 4 is negative and significant at the 0.1% level. This implies that the overall similarity in net job creation rates by earnings inequality is due to two offsetting effects: firms with higher earnings inequality are less likely to exit, but they are also less likely to enter or to grow.

Taken together, our results are consistent with research showing that differences between employers may be a powerful contributing factor to inequality in labor market outcomes for different workers. We find that the differences in average firm net job creation rates and establishment exit rates by firm workforce average age, female share, foreign-born share, or non-white or Hispanic share are meaningful and persistent. Moreover, these differences become *larger* when we compare firms in the same industry or the same MSA. In future work, we plan to produce a BDS-Human Capital experimental product to more comprehensively characterize the firm environments that different workers face in the labor market.

7 Conclusion

The BDS Single-Unit Firms experimental product augments the main BDS in important ways that improve the measurement of economic activity. The most important addition is quarterly data which allow more precise measurement of firm entry and exit and ensure that within-year employment fluctuations are captured. Another important addition is payroll which, as seen in the later half of 2020, sometimes exhibits different dynamics than employment. Similar to the main BDS, the great strength of the published BDS-SU statistics is the rich heterogeneity created by stratifying the tables by firm size, firm age, industry, and geography. This paper provides a summary of much of the information contained in the industry and geography tables but interested readers are encouraged to further investigate tables such as the firm size by firm age by industrial sector to obtain additional detail.

Our analysis of the second through fourth quarters of 2020 provides a detailed picture of the economic contraction caused by the onset of the COVID-19 pandemic. The BDS-SU data show that the pandemic affected the economy in an unequal and heterogeneous manner. The in-person services sectors, including the arts and restaurants, bore the brunt of the decline in overall business activity. In contrast, while the high-tech sectors created jobs and raised wages at a slower pace on net during the pandemic, these metrics never actually declined for firms

in these industries. The variation in the pandemic impact is most evident in differences across states and MSAs in exit and job destruction rates. Some MSAs had exit rates above 20% while others were below 10%.

Differences in firm outcomes carried through to workers. We found that different types of workers were exposed to systematically different economic environments through their employers. Young, female, foreign-born, and non-white or Hispanic workers were especially likely to be working at firms that experienced net job losses or establishment closures in 2020. Furthermore, differences among worker demographic groups in firm net job creation and establishment exit rates were larger when we compared firms in the same industry or the same MSA.

While the pandemic sparked record levels of firm exit, it also ignited a surge in firm entry. In particular, new business applications reached all-time highs, but not in the states and sectors that were hardest hit by the pandemic. Instead, it was the states that remained isolated from the virus in the early months that saw the largest increases in new business ventures. By the end of 2020, new employer businesses appeared at a higher rate, growing more rapidly in the South and West than in the Northeast or Pacific Northwest, the traditional homes of entrepreneurship.

Our analysis in this paper highlights differences in how the pandemic was experienced locally. In some metro areas, circumstances made it difficult for firms to operate whereas in others, conditions enabled businesses to remain open. The authors of this research remain agnostic about what precisely caused different local conditions. Certainly many factors contributed including disease prevelance and severity, local shut-down orders, and changes in consumer behavior. The goal of the BDS-SU product is to measure what happened to businesses during 2020 and provide these data to users evaluating this time period in local areas.

Perhaps the greatest promise of the BDS-SU experimental product is the ability to link to additional data. Research at the Census Bureau is underway to link PPP loan information from the Small Business Administration (SBA) to the LBD in order to tabulate separate exit rates for firms who received loans and grants and those who did not. Efforts will continue to link workers to firms in order to provide information on the demographic profile of different types of firms and the relation to firm dynamics. Finally administrative firm revenue data for single-unit firms are currently being tabulated for a spring 2023 release. These additional tables will measure business dynamics along another dimension, namely annual revenue as reported to the IRS, and will add to our understanding of how small firms were affected by the pandemic.

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Appendix A Codebook and Glossary

The 2020 release of the BDS-SU includes several payroll statistics in both real and nominal terms. Data users can download files with inflation-adjusted payroll variables (real) or non-inflation-adjusted payroll variables (nominal). Each file will contain the standard set of establishment and employment variables outlined below. The inflation-adjusted file names and variable names will have the "_real" suffix. The non-inflation-adjusted file names will have the "_nominal" suffix; note that there is no suffix attached to nominal payroll variable names. To construct real payroll measures, the GDP implicit price deflator from the BEA is used. Note that the base year is 2012.

BDS-SU Variable Definitions and Codes

year
Values: 2007-2020
Description: record year

quarter
Values: 1-4
Description: record quarter

sector
Values: 11,21,22,23,31-33,42,44-45,48-49,51,52,53,54,55,56,61,62,71,72,81
Description: 2-digit NAICS 2017 codes
11 - Agriculture, Forestry, Fishing and Hunting
21 - Mining, Quarrying, and Oil and Gas Extraction
22 - Utilities
23 - Construction
31-33 - Manufacturing
42 - Wholesale Trade
44-45 - Retail Trade
48-49 - Transportation and Warehousing
51 - Information
52 - Finance and Insurance
53 - Real Estate and Rental and Leasing
54 - Professional, Scientific, and Technical Services
55 - Management of Companies and Enterprises
56 - Administrative and Support and Waste Management and Remediation Services
61 - Educational Services
62 - Health Care and Social Assistance
71 - Arts, Entertainment, and Recreation
72 - Accommodation and Food Services
81 - Other Services (except Public Administration)

`vcnaics3`
Values: Numeric
Description: Vintage-consistent 3-digit NAICS 2017 codes.

`vcnaics4`
Values: Numeric
Description: Vintage-consistent 4-digit NAICS 2017 codes.

`fsize`
Values: a) 1 to 2, b) 3 to 9, c) 10 to 99, d) 100 to 499, e) 500+
Description: Firm (establishment) size. A categorical variable defined as the average of the single-establishment firm's employment in quarter qt and $qt-4$.

`fsizecoarse`
Values: a) 1 to 2, b) 3 to 9, c) 10+
Description: Firm (establishment) size coarse. This is the coarse version of fsize defined above. Includes three broader firm-size classes.

`fage`
Values: a) 0, b) 1 to 3, c) 4 to 20, d) 21 to 40, e) 41+, f) Left Censored
Description: The number of calendar quarters that have elapsed since a single-establishment firm's first quarter of operation.

`fagecoarse`
Values: a) 0 to 3, b) 4 to 20, c) 21 to 40, d) 41+, e) Left Censored
Description: Firm age coarse with five broad firm-age classes.

`estabs`
Values: Numeric
Description: A simple count of the number of single-establishment firms in the cell.

`emp`
Values: Numeric
Description: Paid employment consists of full- and part-time employees, including salaried officers and executives of corporations, who are on the payroll in the pay periods including March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$). Included are employees on paid sick leave, holidays, and vacations.

`denom`
Values: Numeric
Description: Davis-Haltiwanger-Schuh (DHS) denominator. For quarter qt, `denom` is the average of employment in quarters qt and $qt-4$. This variable attempts to prevent transitory shocks from creating a bias to the relationship between net growth from $qt-4$ to qt and size.

`estabs_entry`
Values: Numeric
Description: A count of single-establishment firms born within the cell during the previous four

quarters.

estabs_entry_rate
Values: Numeric
Description: 100 x (estabs_entry in quarter qt divided by the average of estabs in quarters qt and $qt-4$)

estabs_exit
Values: Numeric
Description: A count of single-establishment firms exiting from within the cell during the previous four quarters.

estabs_exit_rate
Values: Numeric
Description: 100 x (estabs_exit in quarter qt divided by the average of estabs in quarters qt and $qt-4$)

job_creation
Values: Numeric
Description: A count of all employment gains within the cell occurring over the previous four quarters from expanding and opening single-establishment firms. The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

job_creation_births
Values: Numeric
Description: A count of all employment gains within the cell occurring over the previous four quarters from opening single-establishment firms (births). The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

job_creation_continuers
Values: Numeric
Description: A count of all employment gains within the cell occurring over the previous four quarters from continuing single-establishment firms. The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

job_creation_rate_births
Values: Numeric
Description: 100 x (job_creation_births/denom)

job_creation_rate
Values: Numeric
Description: 100 x (job_creation/denom)

job_destruction
Values: Numeric
Description: A count of all employment losses within the cell occurring over the previous four quarters from contracting and closing single-establishment firms. The reference weeks are as

follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

job_destruction_deaths
Values: Numeric
Description: A count of all employment losses within the cell occurring over the previous four quarters from closing single-establishment firms (deaths). The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

job_destruction_continuers
Values: Numeric
Description: A count of all employment losses within the cell occurring over the previous four quarters from contracting single-establishment firms. The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

job_destruction_rate_deaths
Values: Numeric
Description: 100 x (job_destruction_deaths/denom)

job_destruction_rate
Values: Numeric
Description: 100 x (job_destruction/denom)

net_job_creation
Values: Numeric
Description: job_creation - job_destruction

net_job_creation_rate
Values: Numeric
Description: job_creation_rate - job_destruction_rate

pay
Values: Numeric
Description: Payroll (in thousands of dollars) at single-establishment firms with positive employment during the pay periods including March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$). Included are employees on paid sick leave, holidays, and vacations.

denom_pay
Values: Numeric
Description: The average of payroll in quarters qt and $qt-4$ among single-establishment firms with positive employment. This variable is analogous to denom for employment.

pay_creation
Values: Numeric
Description: The dollar value of payroll gains (in thousands of dollars) occurring over the previous four quarters from expanding and opening single-establishment firms. The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

pay_creation_births
Values: Numeric
Description: The dollar value of payroll gains (in thousands of dollars) occurring over the previous four quarters from opening single-establishment firms (births). The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

pay_creation_continuers
Values: Numeric
Description: The dollar value of payroll gains (in thousands of dollars) occurring over the previous four quarters from expanding single-establishment firms. The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

pay_creation_rate_births
Values: Numeric
Description: 100 x (pay_creation_births/denom_pay)

pay_creation_rate
Values: Numeric
Description: 100 x (pay_creation/denom_pay)

pay_destruction
Values: Numeric
Description: The dollar value of payroll losses (in thousands of dollars) occurring over the previous four quarters from contracting and closing single-establishment firms. The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

pay_destruction_deaths
Values: Numeric
Description: The dollar value of payroll losses (in thousands of dollars) occurring over the previous four quarters from closing single-establishment firms (deaths). The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

pay_destruction_continuers
Values: Numeric
Description: The dollar value of payroll losses (in thousands of dollars) occurring over the previous four quarters from contracting single-establishment firms that contracted. The reference weeks are as follows: March 12 ($q1$), June 12 ($q2$), September 12 ($q3$), and December 12 ($q4$).

pay_destruction_rate_deaths
Values: Numeric
Description: 100 x (pay_destruction_deaths/denom_pay)

pay_destruction_rate
Values: Numeric
Description: 100 x (pay_destruction/denom_pay)

net_pay_creation

Values: Numeric
Description: pay_creation - pay_destruction

net_pay_creation_rate
Values: Numeric
Description: pay_creation_rate - pay_destruction_rate

pay_emp
Values: Numeric
Description: Payroll per worker at single-establishment firms (in thousands of dollars).

pay_emp_continuers
Values: Numeric
Description: Payroll per worker at continuing single-establishment firms (in thousands of dollars).

pay_emp_births
Values: Numeric
Description: Payroll per worker at opening single-establishment firms (in thousands of dollars).

pay_emp_deaths
Values: Numeric
Description: Payroll per worker at closing single-establishment firms (in thousands of dollars).

metro
Values: M,N,SW,U
Description: Single-establishment firms located in Metropolitan or Micropolitan Statistical Areas are in cells designated with "M" while those in non-metropolitan areas are aggregated into cells designated with "N". Cells designated "SW" (signifying "statewide") are assigned to single-establishment firms without a fixed location such as employee leasing establishments. Cells designated as "U" (signifying "unclassified") are assigned to single-establishment firms without a valid county code.

msac
Values: Numeric
Description: Numeric code for metropolitan and micropolitan statistical areas, as established by the Office of Management and Budget (OMB). This is a coarse msa identifier in that micropolitan statistical areas are grouped together under the identifier, "Micro". The February 2013 definitions are used: https://www.census.gov/geographies/reference-files/time-series/demo/metro-micro/delineation-files.html

cty
Values: Numeric
Description: Three-character FIPS code for counties within a given state. Users can refer to the CBP county reference files for listings of valid county codes here: https://www.census.gov/programs-surveys/cbp/technical-documentation/reference/state-county-geography-reference.html. The BDS-SU also includes statewide (999) and unclassified (998) county codes. Statewide

county codes are assigned to single-establishment firms without a fixed location such as employee leasing establishments. Unclassified single-establishment firms are those with a county code that cannot be found in any of the CBP county reference files.

st
Values: 01,02,04,05,06,08,09,10,11,12,13,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,44,45,46,47,48,49,50,51,53,54,55,56
Description: 2-digit FIPS codes
01 – Alabama (AL)
02 – Alaska (AK)
04 – Arizona (AZ)
05 – Arkansas (AR)
06 – California (CA)
08 – Colorado (CO)
09 – Connecticut (CT)
10 – Delaware (DE)
11 – District of Columbia (DC)
12 – Florida (FL)
13 – Georgia (GA)
15 – Hawaii (HI)
16 – Idaho (ID)
17 – Illinois (IL)
18 – Indiana (IN)
19 – Iowa (IA)
20 – Kansas (KS)
21 – Kentucky (KY)
22 – Louisiana (LA)
23 – Maine (ME)
24 – Maryland (MD)
25 – Massachusetts (MA)
26 – Michigan (MI)
27 – Minnesota (MN)
28 – Mississippi (MS)
29 – Missouri (MO)
30 – Montana (MT)
31 – Nebraska (NE)
32 – Nevada (NV)
33 – New Hampshire (NH)
34 – New Jersey (NJ)
35 – New Mexico (NM)
36 – New York (NY)
37 – North Carolina (NC)
38 – North Dakota (ND)
39 – Ohio (OH)
40 – Oklahoma (OK)
41 – Oregon (OR)
42 – Pennsylvania (PA)

44 – Rhode Island (RI)
45 – South Carolina (SC)
46 – South Dakota (SD)
47 – Tennessee (TN)
48 – Texas (TX)
49 – Utah (UT)
50 – Vermont (VT)
51 – Virginia (VA)
53 – Washington (WA)
54 – West Virginia (WV)
55 – Wisconsin (WI)
56 – Wyoming (WY)

BDS-SU Glossary

Business Dynamics Statistics (BDS) – A public dataset on the LBD that describes United States business dynamics across a wide range of measures. Disclosure analysis is performed prior to release to the public to protect the confidentiality of the underlying LBD data.

Business Register (BR) – A comprehensive database of all U.S. business establishments developed and maintained by the U.S. Census Bureau, with data beginning in 1975 and continuing to the present. This is restricted data and is the source data for the Longitudinal Business Database (LBD).

Censoring – A statistical term indicating that a value cannot be known with certainty. Within the BDS, all firms and establishments born prior to 1976 have an unknown birth year and are therefore of an unknown age and are grouped into the age category "Left Censored".

Data Quality Suppression – Data quality suppressions are made when a cell is determined to be unreliable due to its time series characteristics. Cells suppressed due to data quality concerns will appear as "(S)".

Disclosure Suppression – Disclosure suppressions are made when a cell has too few firms. Cells suppressed due to containing too few firms will appear as "(D)".

Gross Domestic Product (GDP) Implicit Price Deflator – The price index produced by the Bureau of Economic Analysis (BEA), which is used to construct inflation-adjusted (real) payroll measures. The data can be found on FRED here: https://fred.stlouisfed.org/series/GDPDEF. In using this deflator, we maintain the 2012 base year.

Nominal Variables – Data that are unadjusted for inflation. Nominal data series are in "current" dollar terms.

Real Variables – Data that are adjusted for inflation. Real data series are in "constant" dollar terms.

Structurally Missing Flag – Structurally missing flags are applied to cells that are "structurally zero" or "structurally missing." These are cells in the firm and establishment datasets where activity is not possible given the nature or structure of the BDS data. These cells appear as "(X)". An example of a "structurally zero" cell is the variable `estabs_exit` for firms age '0'. These cells will always be '0' given the nature of data for firms age '0'. An example of a "structurally missing" cell is any variable for firms age '5' in the years 1978 to 1981. These cells will always be missing for the years 1978 to 1981 because the source data for the BDS – the LBD – begins in 1976.

North American Industrial Classification System (NAICS) – NAICS is the standard used by Federal statistical agencies in classifying business establishments for the purposes of collecting, analyzing, and publishing statistical data released to the U.S. business economy. The system is used by the United States, Canada, and Mexico. Note that the 2020 BDS-SU release I based on the 2017 NAICS vintage.

Appendix B Figures

Figure 1: Employment Time-Series in the BDS-SU: 2007-2020

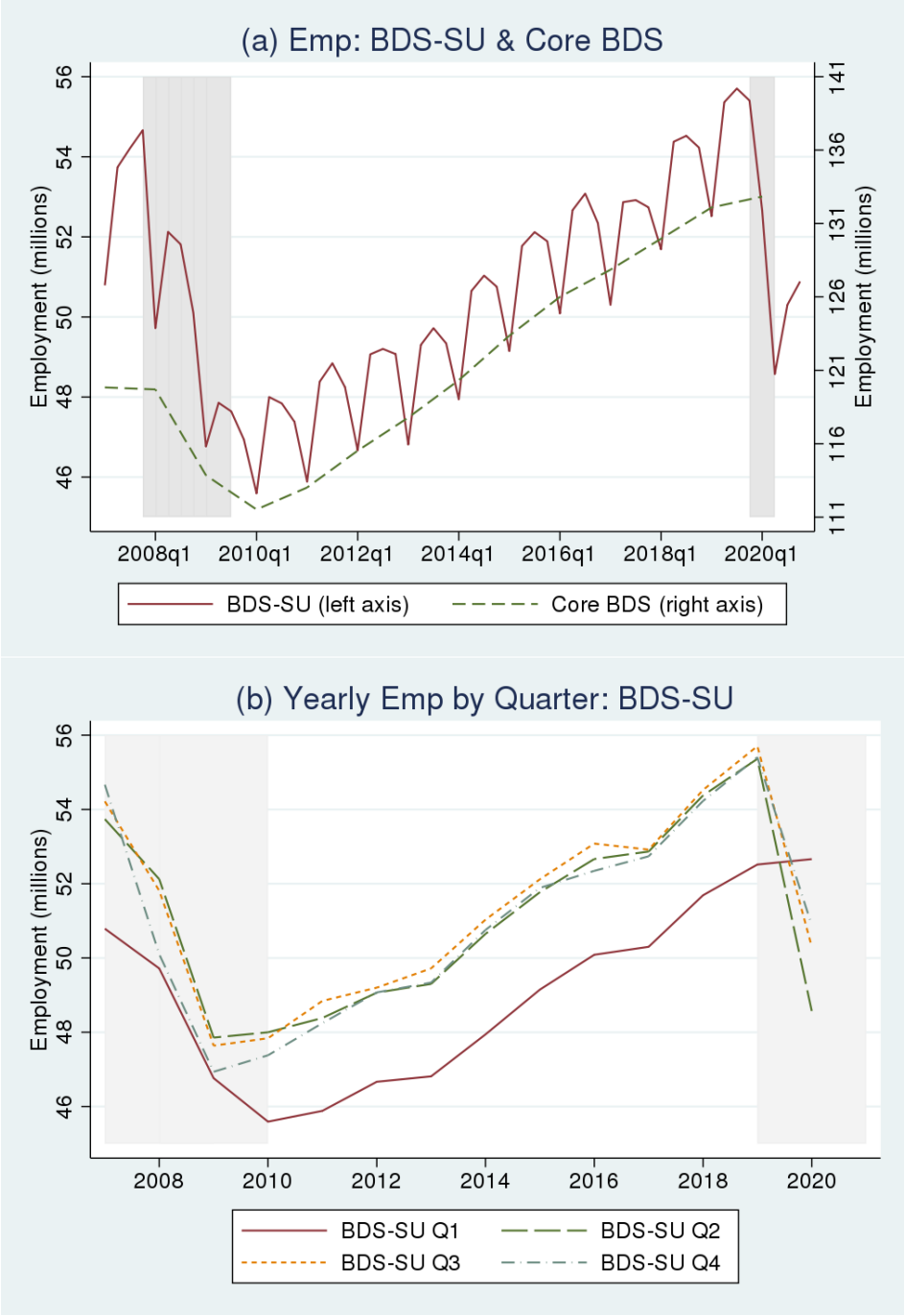


Figure 2: Job Creation in BDS-SU & Core BDS: 2007-2020

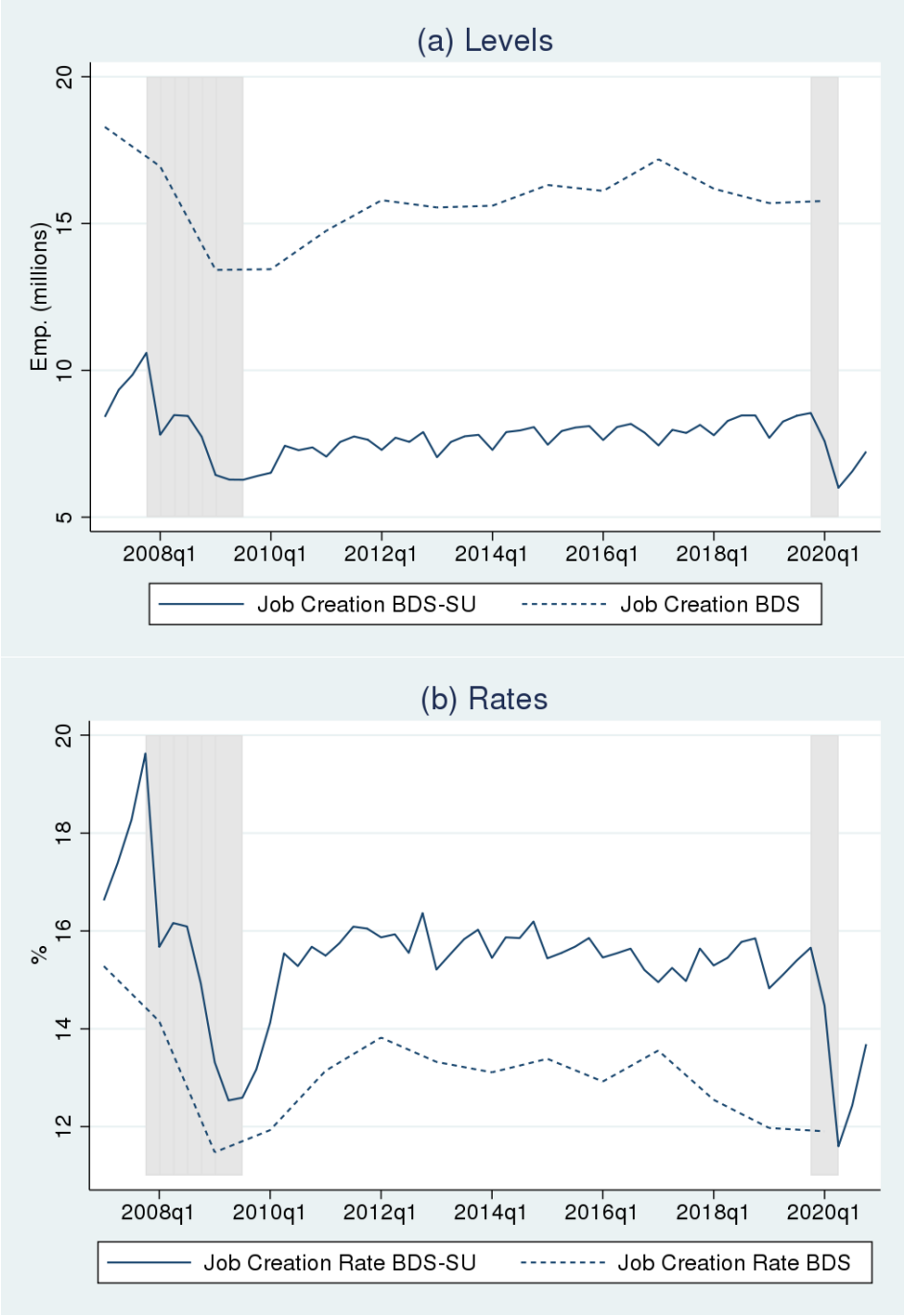


Figure 3: Job Destruction in BDS-SU & Core BDS: 2007-2020

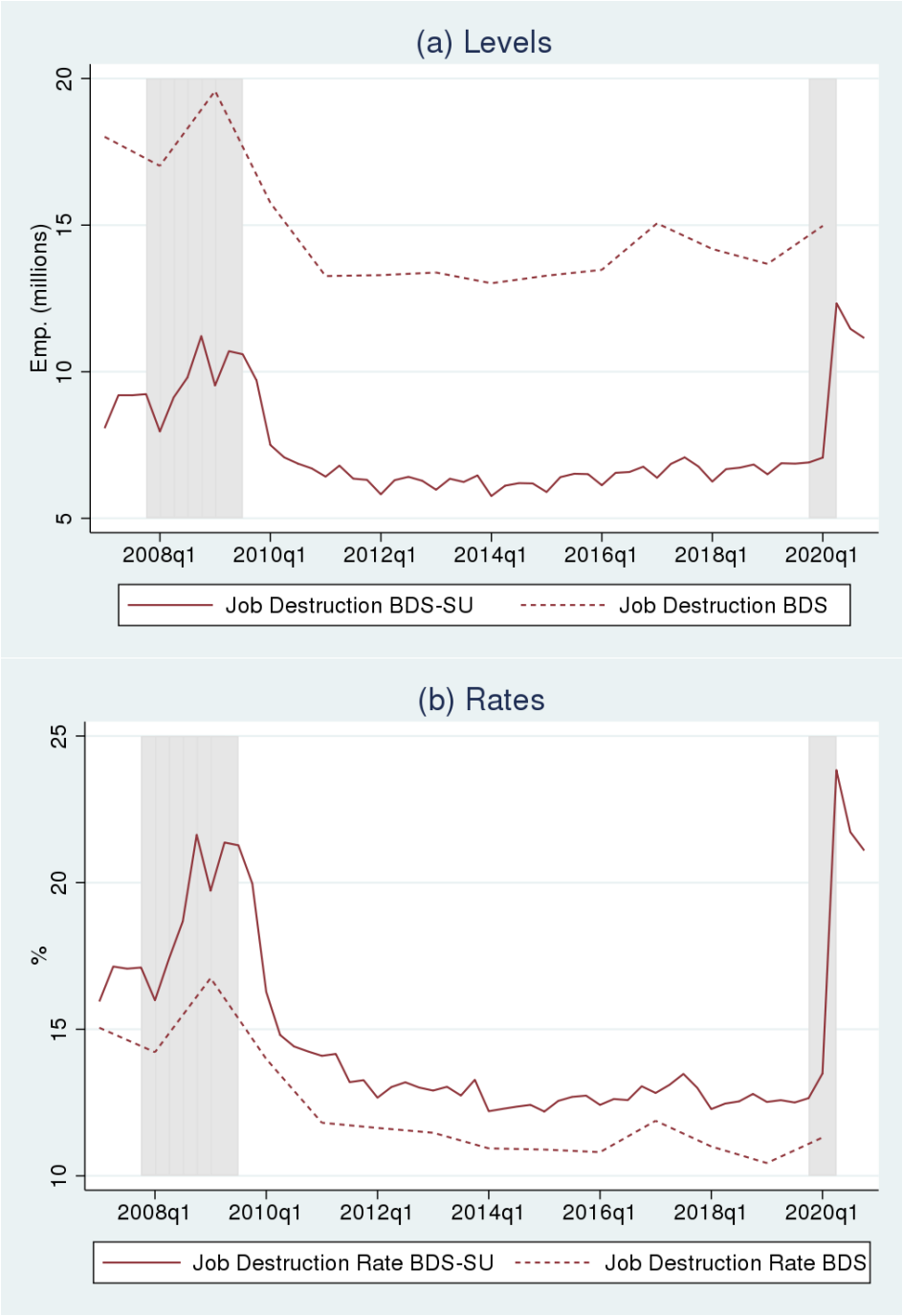
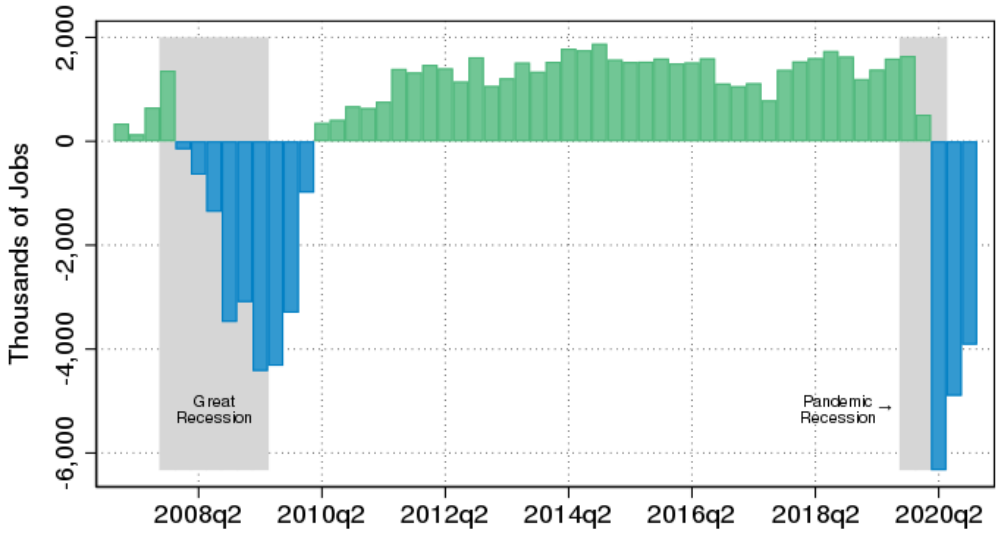
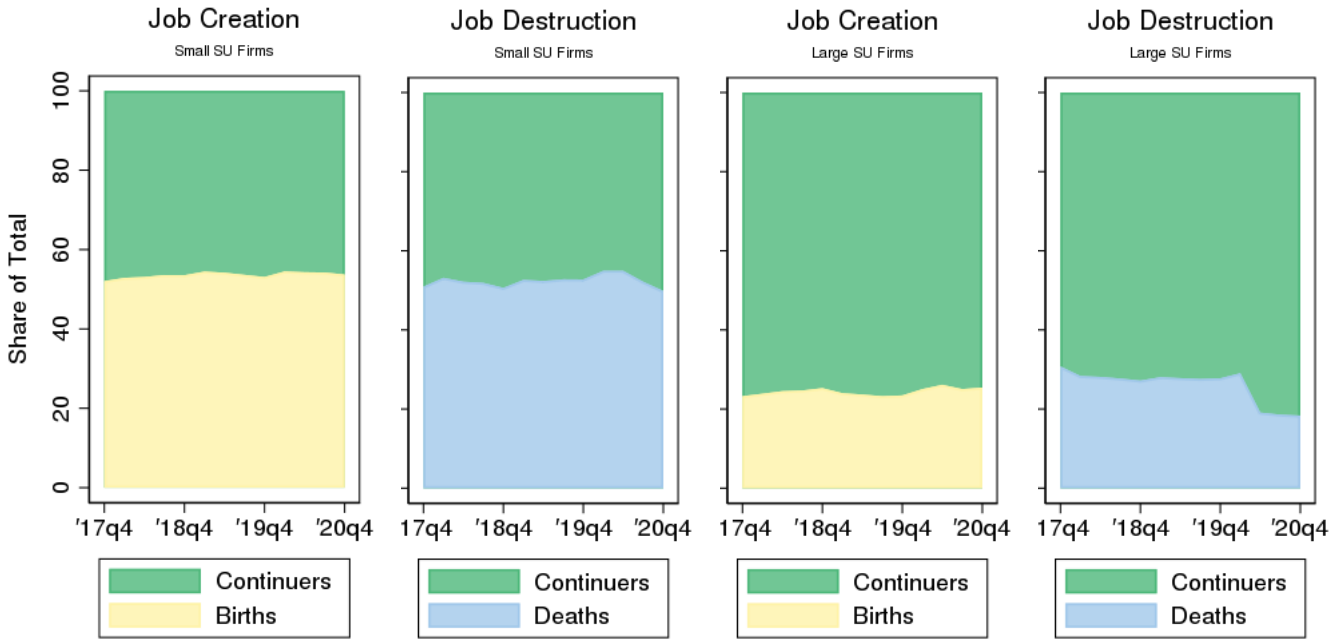


Figure 4: Net Job Creation for Single-Unit Firms



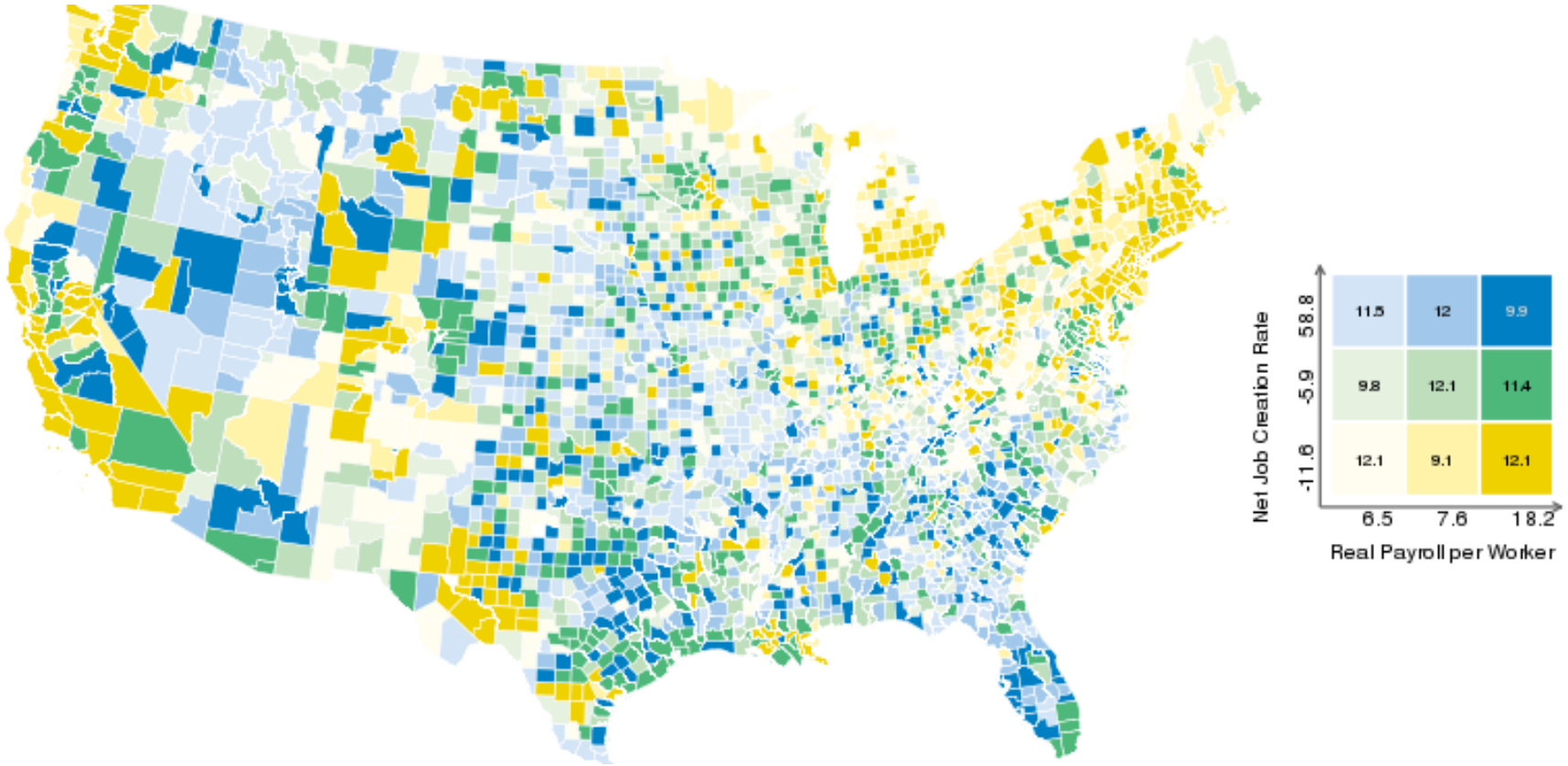
Note: This figure shows net job creation for single-unit firms starting in 2007q1 and ending in the most-recent quarter, 2020q4. NBER-defined recessions shaded in gray.

Figure 5: Job Creation and Destruction Shares by Firm Size during the Pandemic



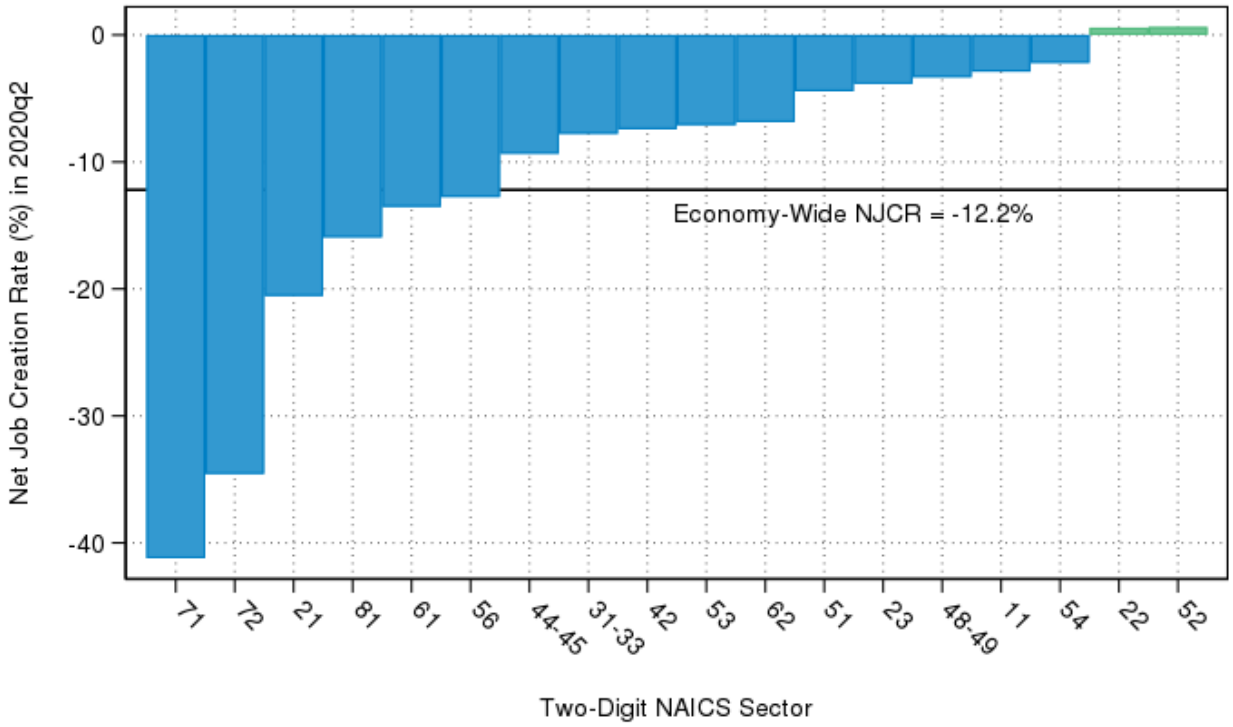
Note: This figure shows how continuing single-unit firms and new entrants (births) contribute to job creation. It also shows how continuers and exits (deaths) contribute to job destruction, especially during the pandemic. Small single-unit firms employ less than 10 workers, and large single-unit firms employ 10 or more workers. By definition, deaths cannot contribute to job creation just as births cannot contribute to job destruction.

Figure 6: Real Payroll-per-Worker and Net Job Destruction Rates across U.S. Counties



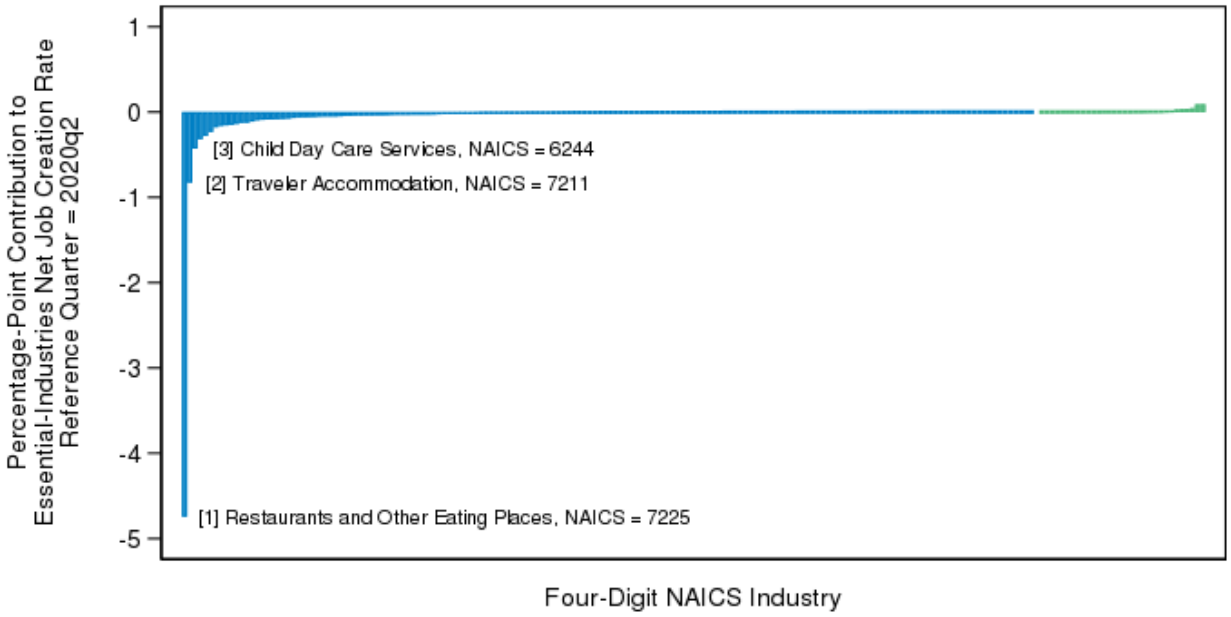
Note: This bivariate map aims to understand the income characteristics (real payroll-per-worker) of U.S. counties that experienced various degrees of net job creation during the pandemic. Real payroll-per-worker in 2019q2 is plotted against the net job creation rate in 2020q2. Along each axis, each bin represents a tercile of the data. The values marked along the horizontal and vertical axis represent an upper bound, or "cut-off", for each bin. The number presented inside each bin denotes the percentage of counties that fall within that bin. For example, the bin in the bottom left corner of the legend (light gold) includes counties that were in the bottom third in terms of both real payroll-per-worker and the net job creation rate. The bin to the bottom right (dark gold), includes the 12.1 percent of all counties that recorded high payroll-per-worker (top third) but also reported the most negative net job creation rates.

Figure 7: Net Job Creation Rate by Sector



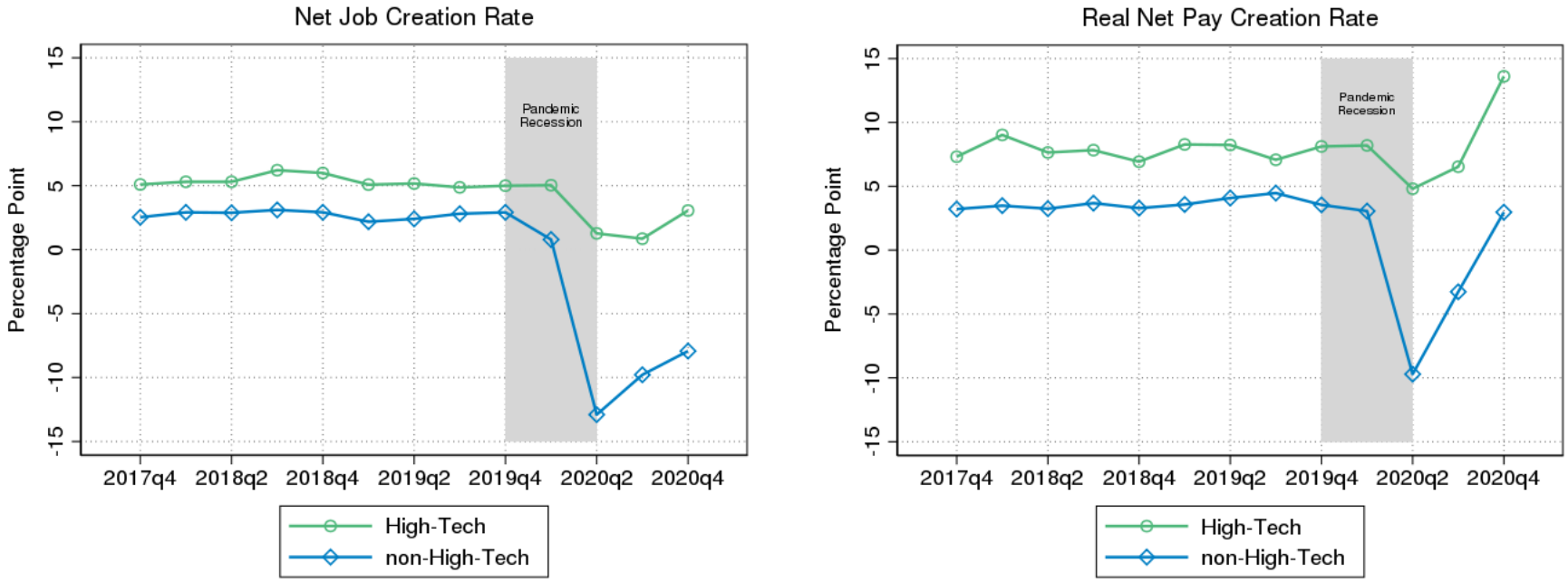
Note: This figure plots each sector's net job creation rate in 2020q2 when the national rate fell to its trough during the pandemic. For comparison, the national decline in the net job creation rate in 2020q2 measured -12.2 percent. Omitted from this graph sectors 55 (Management of Companies and Enterprises), which tends to exhibit large variation not attributable to underlying economic conditions, and 92 (Public Administration), which is out of scope in the BDS-SU. NAICS sectors: 71 (Arts, Entertainment, and Recreation), 72 (Accommodation and Food Services), 21 (Mining, Quarrying, and Oil and Gas Extraction), 81 (Other Services), 61 (Educational Services), 56 (Administrative and Support and Waste Management and Remediation Services), 44-45 (Retail Trade), 31-33 (Manufacturing), 42 (Wholesale Trade), 53 (Real Estate and Rental and Leasing), 62 (Health Care and Social Assistance), 51 (Information), 23 (Construction), 48-49 (Transportation and Warehousing), 11 (Agriculture, Forestry, Fishing and Hunting), 54 (Professional, Scientific, and Technical Services), 22 (Utilities), and 52 (Finance and Insurance).

Figure 8: Job Loss at Essential Industries during the Pandemic



Note: Essential industries are defined by Blau, Koebe, and Meyerhofer (2021) and include the following four-digit NAICS codes: 1131, 1132, 1133, 1141, 1142, 1151, 1152, 1153, 2111, 2121, 2122, 2123, 2131, 2211, 2212, 2213, 2361, 2362, 2371, 2372, 2373, 2379, 2381, 2382, 2383, 2389, 3111, 3112, 3113, 3114, 3116, 3117, 3118, 3119, 3121, 3211, 3212, 3219, 3221, 3222, 3231, 3241, 3251, 3252, 3253, 3254, 3255, 3256, 3259, 3261, 3262, 3271, 3272, 3273, 3274, 3279, 3311, 3312, 3313, 3314, 3315, 3321, 3322, 3323, 3324, 3325, 3326, 3327, 3328, 3329, 3331, 3332, 3333, 3334, 3335, 3336, 3339, 3341, 3342, 3343, 3344, 3345, 3346, 3351, 3352, 3353, 3359, 3361, 3362, 3363, 3364, 3365, 3366, 3369, 3391, 4233, 4234, 4235, 4236, 4237, 4238, 4241, 4242, 4244, 4245, 4246, 4247, 4248, 4249, 4251, 4413, 4431, 4441, 4442, 4451, 4452, 4453, 4461, 4471, 4523, 4541, 4543, 4811, 4812, 4831, 4841, 4842, 4851, 4852, 4853, 4854, 4855, 4859, 4861, 4862, 4869, 4881, 4882, 4883, 4884, 4885, 4889, 4921, 4922, 4931, 5111, 5112, 5151, 5152, 5173, 5182, 5191, 5221, 5222, 5223, 5231, 5232, 5239, 5241, 5259, 5311, 5312, 5313, 5321, 5412, 5416, 5417, 5419, 5616, 5617, 5621, 5622, 5629, 6211, 6212, 6213, 6214, 6215, 6216, 6219, 6221, 6222, 6223, 6231, 6232, 6233, 6239, 6241, 6242, 6244, 7211, 7223, 7225, 8111, 8112, 8113, 8122. Net job creation rates for each industry are weighted by their denom share. The overall net job creation rate for essential (non-essential) industries in 2020q2 measured -10.3 percent (-16.6 percent).

Figure 9: Comparing High-Tech and non-High-Tech Sectors



Note: The high-tech designation includes the following four-digit NAICS codes: Computer and Peripheral Equipment Manufacturing (3341), Communications Equipment Manufacturing (3342), Semiconductor and Other Electronic Component Manufacturing (3344), Navigational, Measuring, Electromedical, and Control Instruments Manufacturing (3345), Aerospace Product and Parts Manufacturing (3364), Software Publishers (5512), Data Processing, Hosting, and Related Services (5182), Other Information Services (5191), Architectural, Engineering, and Related Services (5413), Computer Systems Design and Related Services (5415), and Scientific Research and Development Services (5417). For comparison, high-tech single-units represent roughly 95% of total high-tech firms, 76% of total high-tech establishments, and 33% of total high-tech employment during the 2007-2020 period. Within the single-unit universe, high-tech has a relatively small presence, representing roughly 4.2% of all single-unit firms. NBER-defined recession shaded in gray.

Figure 10: Net Pay Creation in BDS-SU: 2007-2020

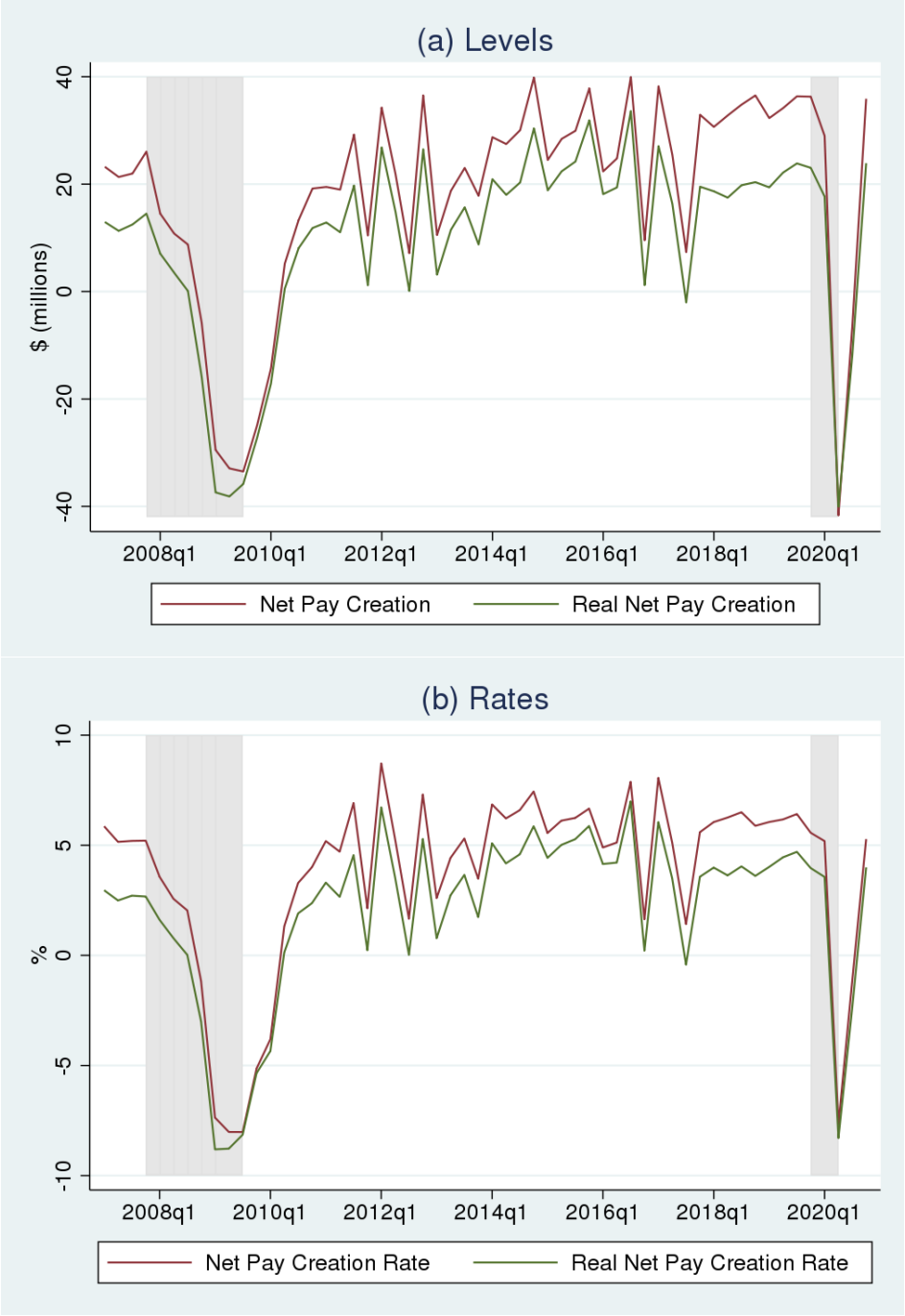


Figure 11: Payroll per Employee in BDS-SU: 2007-2020

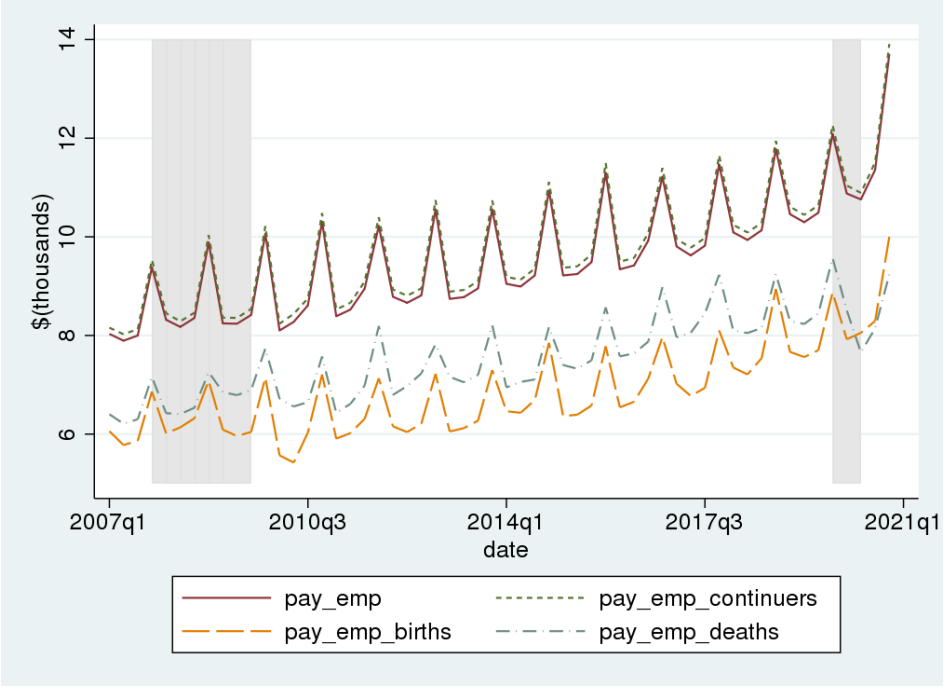
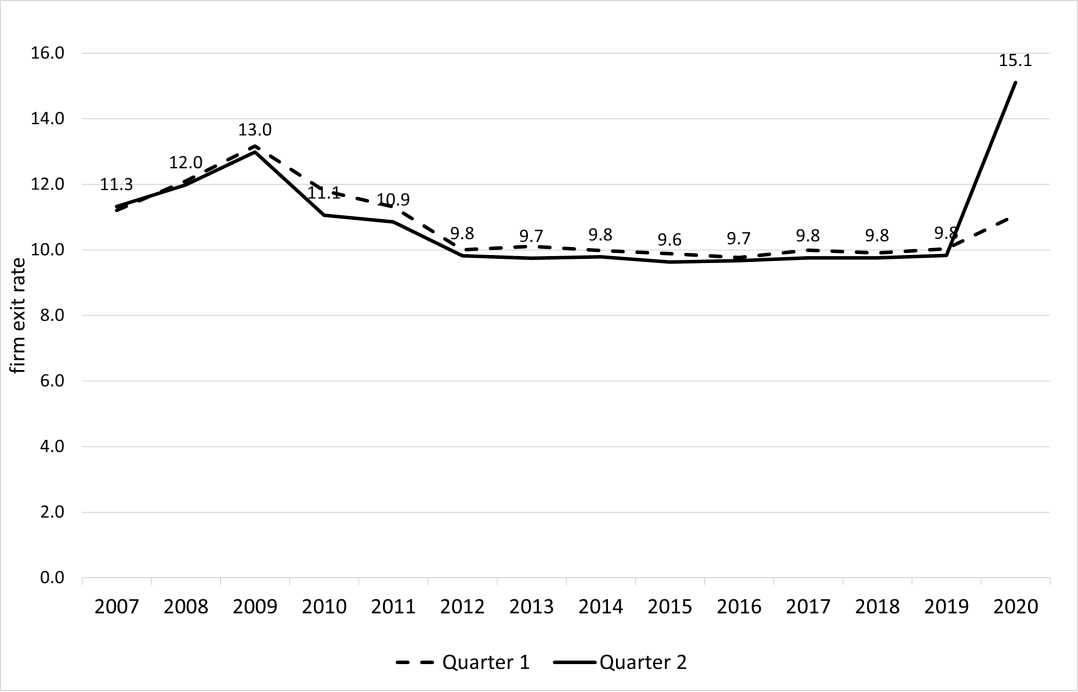
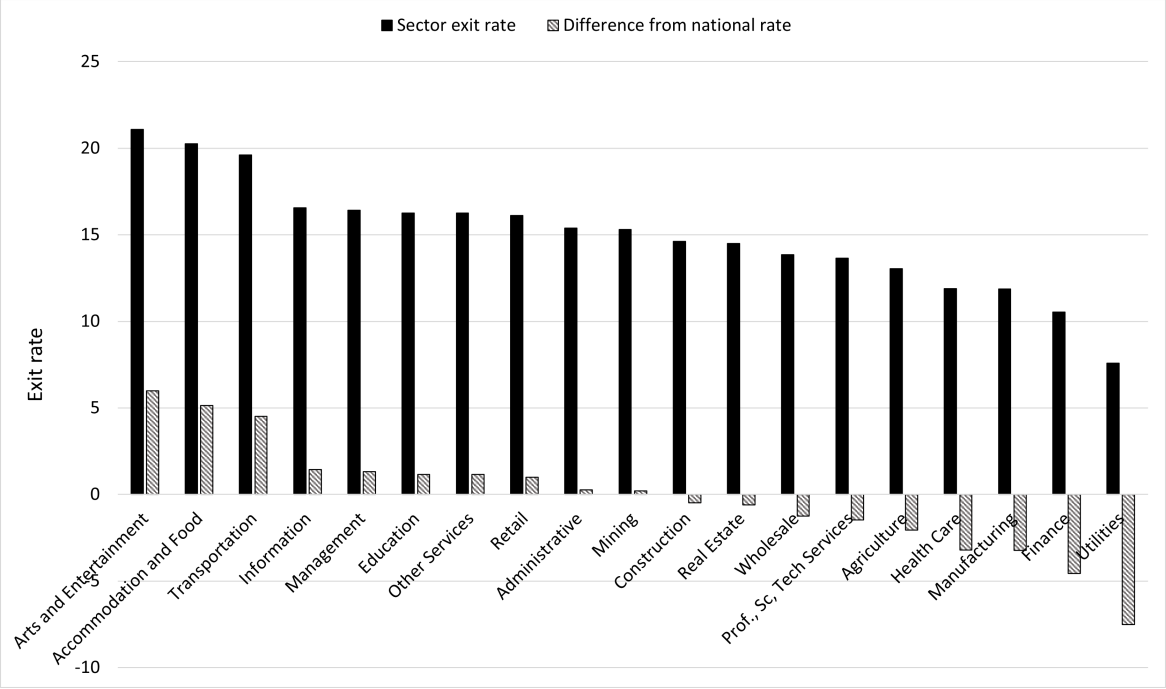


Figure 12: Single-unit firms exit rate: 2007-2020 Quarters 1 and 2



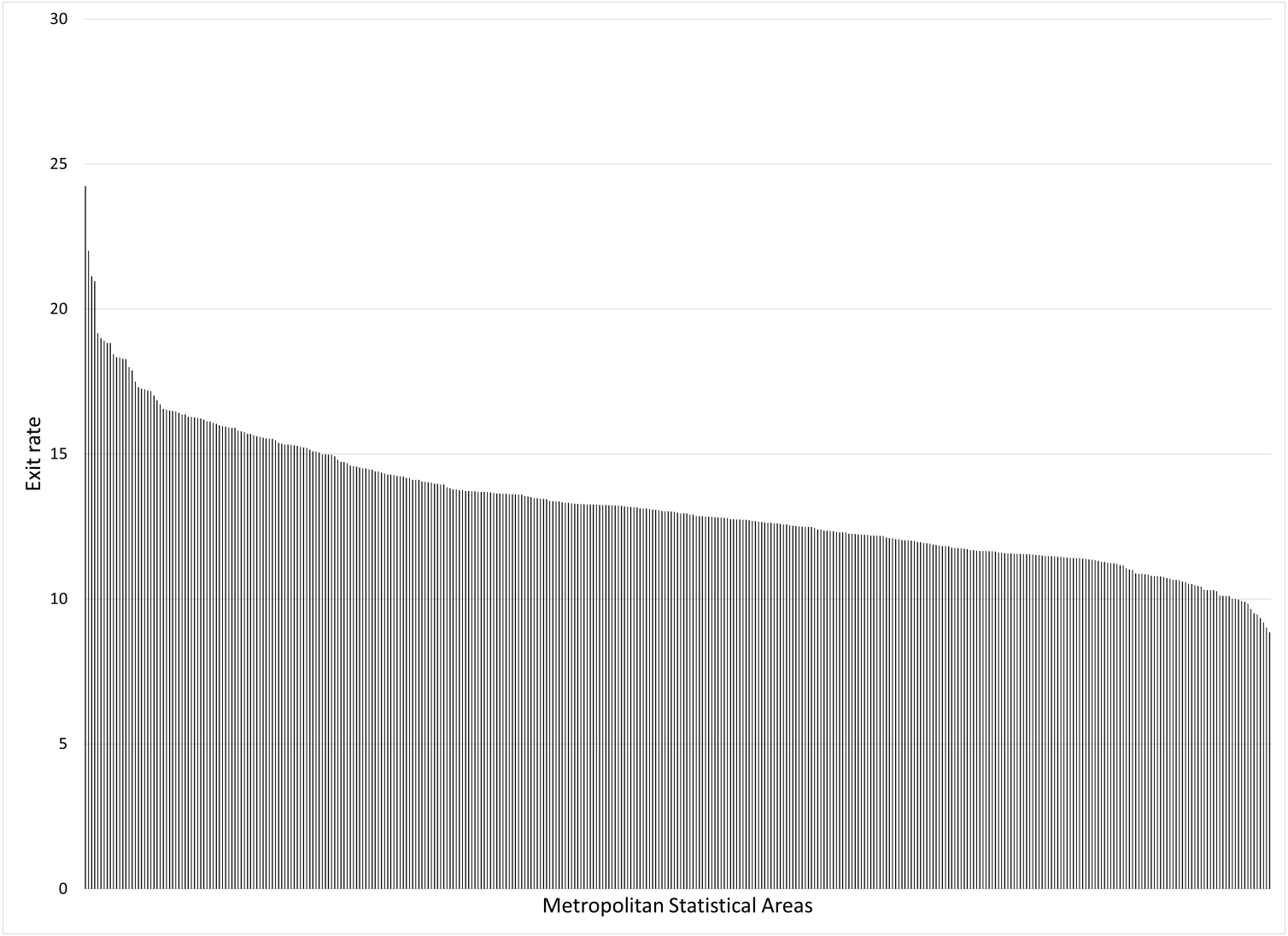
Note: A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 13: Distribution of exit rates across NAICS Sectors: 2020 Quarter 2



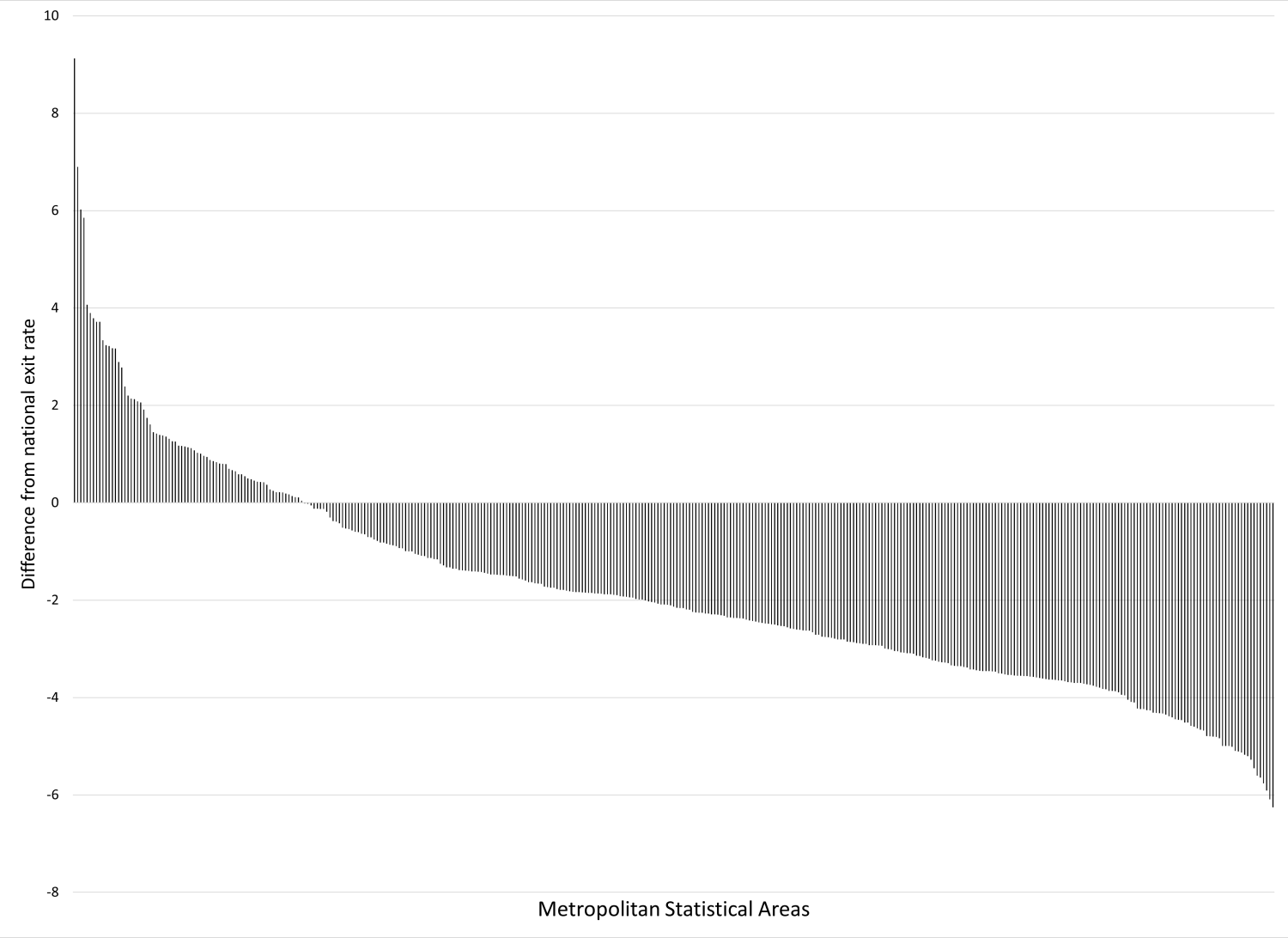
Note: A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Industry sector is based on 2017 NAICS codes from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 14: Distribution of exit rates across Metropolitan Statistical Areas: 2020 Quarter 2



Note: A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 15: MSA differences from national exit rate: 2020 Quarter 2



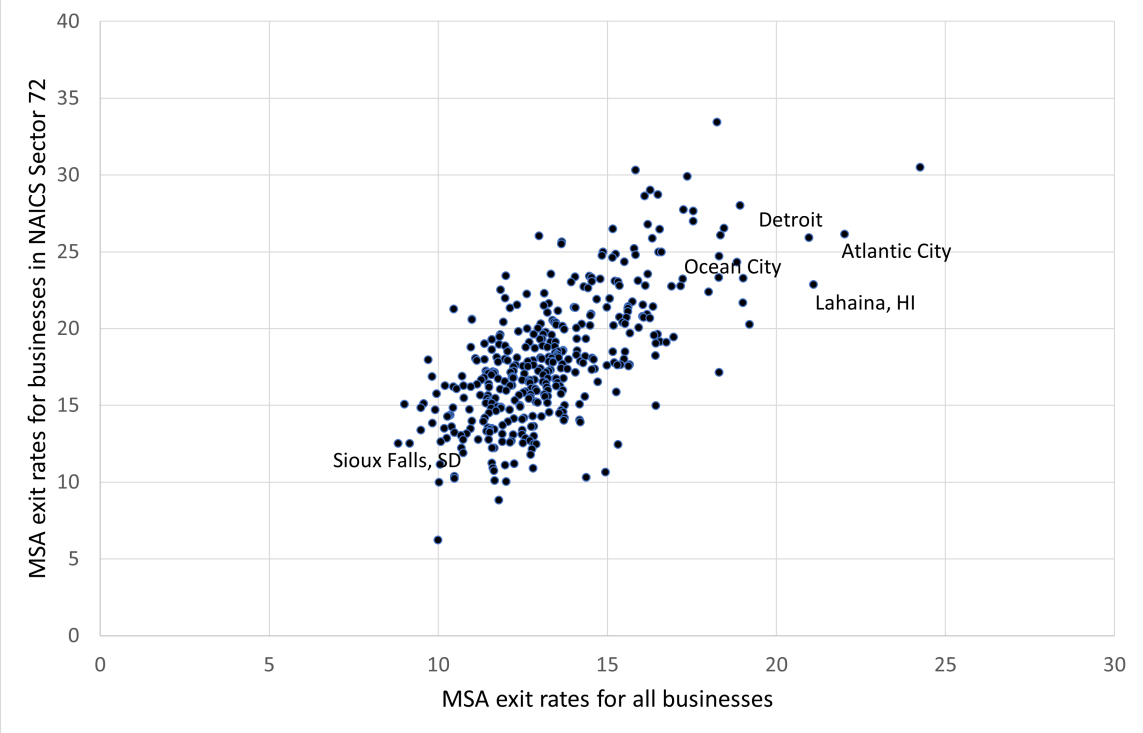
Note: This graph plots the difference between the exit rate for each MSA and the national exit rate of 15.1%. A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 16: MSA exit rates compared to MSA share of hospitality sector: 2020 Quarter 2



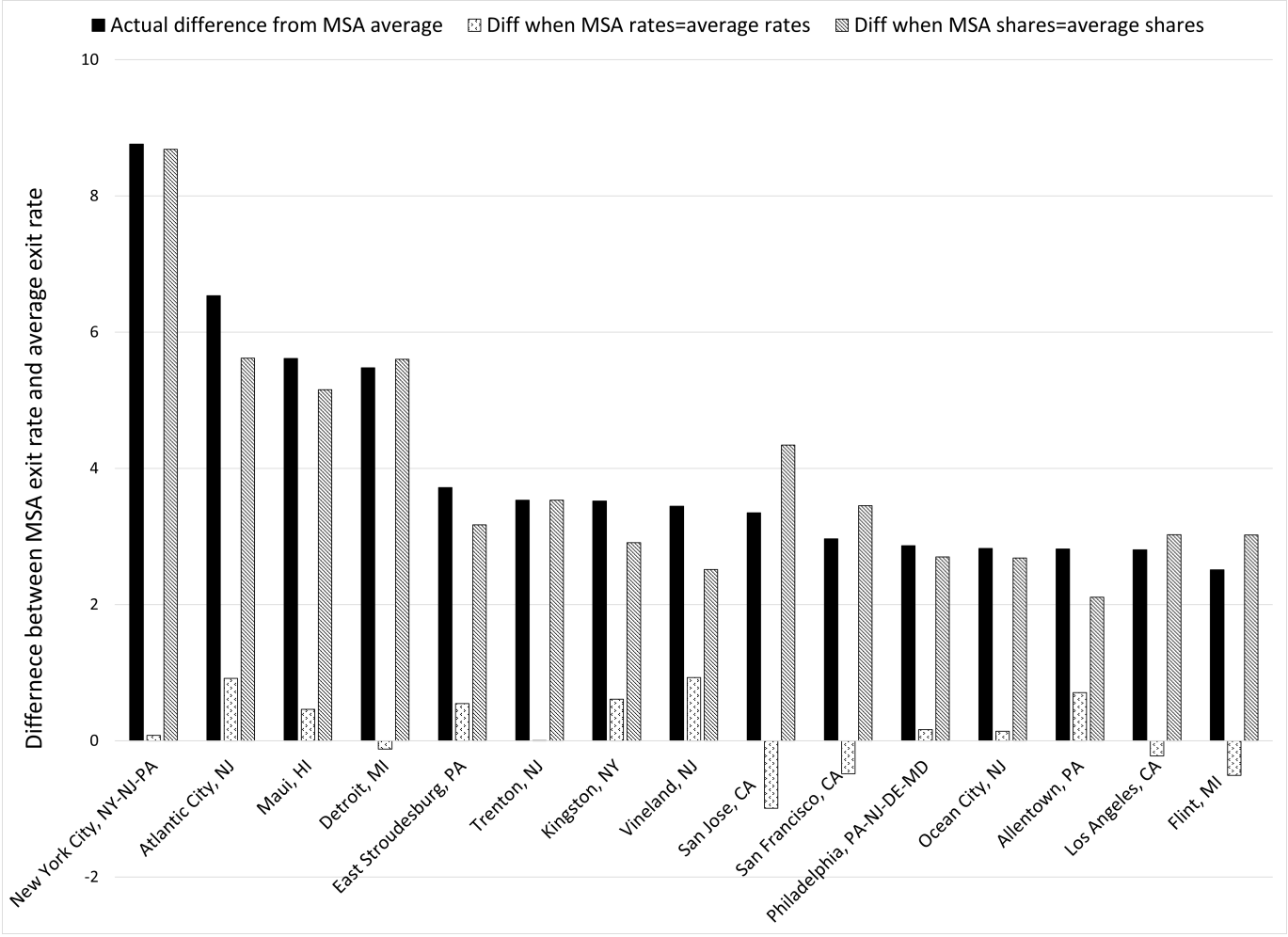
Note: This graph plots a pair of data points for each MSA: (MSA exit rate, share of establishments in MSA in sector 72). A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Industry sector is based on 2017 NAICS codes from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 17: MSA exit rates compared to MSA hospitality sector exit rates: 2020 Quarter 2



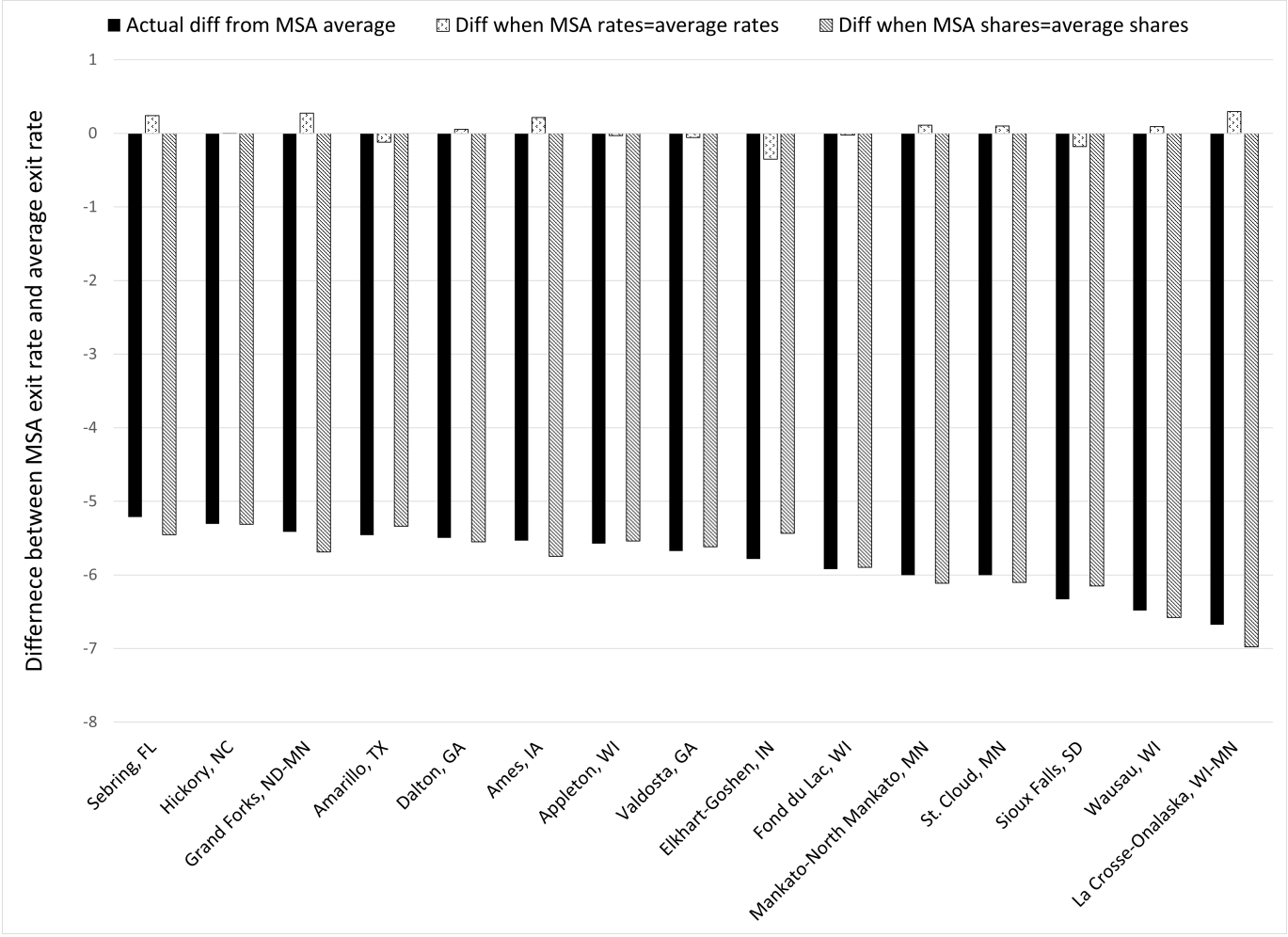
Note: This graph plots a pair of data points for each MSA: (MSA exit rate, MSA sector 72 exit rate). A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Industry sector is based on 2017 NAICS codes from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 18: Effect of MSA industry shares versus industry exit rates for MSAs with top 15 exit rates



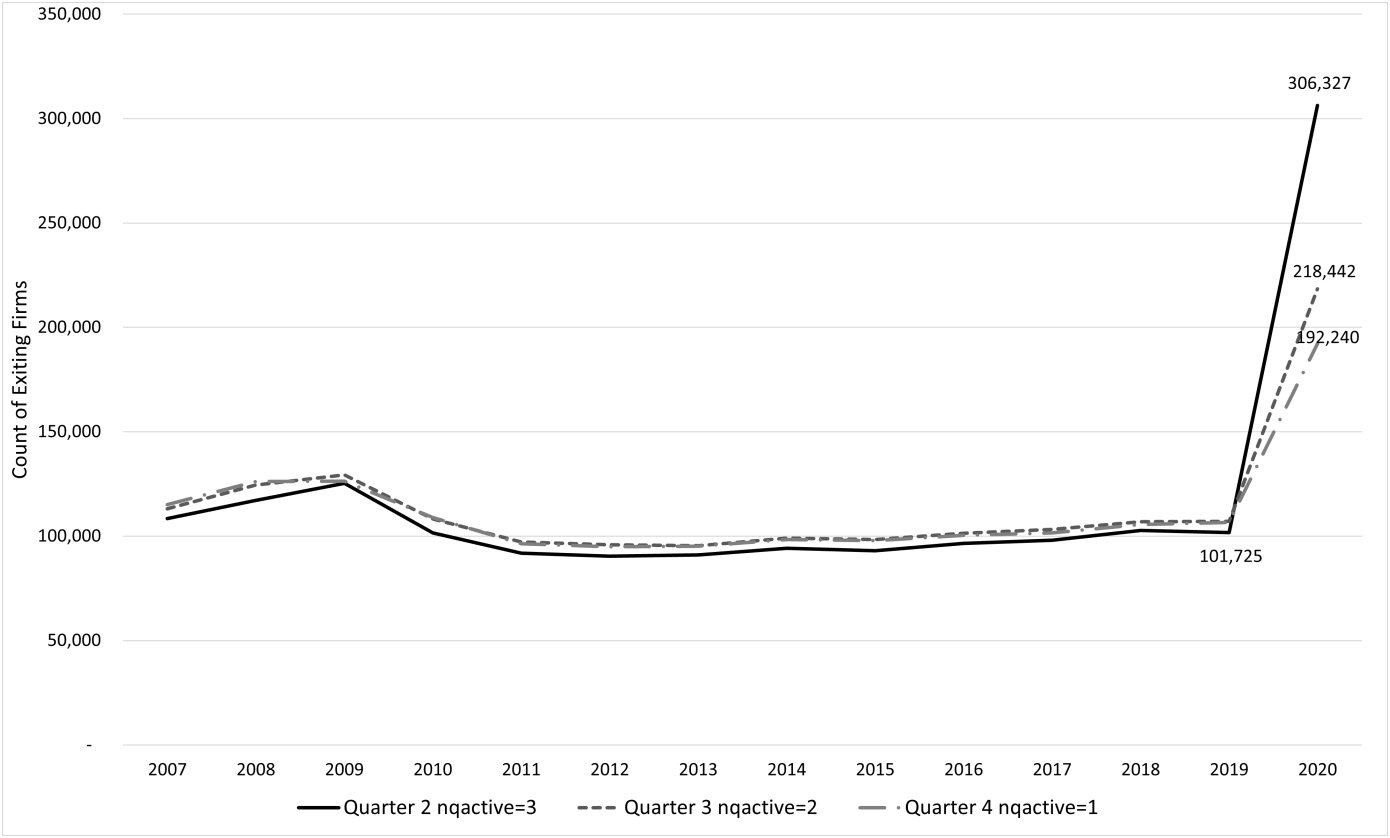
Note: This graph shows the actual difference between the MSA exit rate and the average exit rate across all MSAs for the 15 MSAs with the highest exit rates. It also shows two counterfactual differences: one calculated using MSA-average sector-specific exit rates and one calculated using MSA-average share of firms in each sector. A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 19: Effect of MSA industry shares versus industry exit rates for MSAs with bottom 15 exit rates



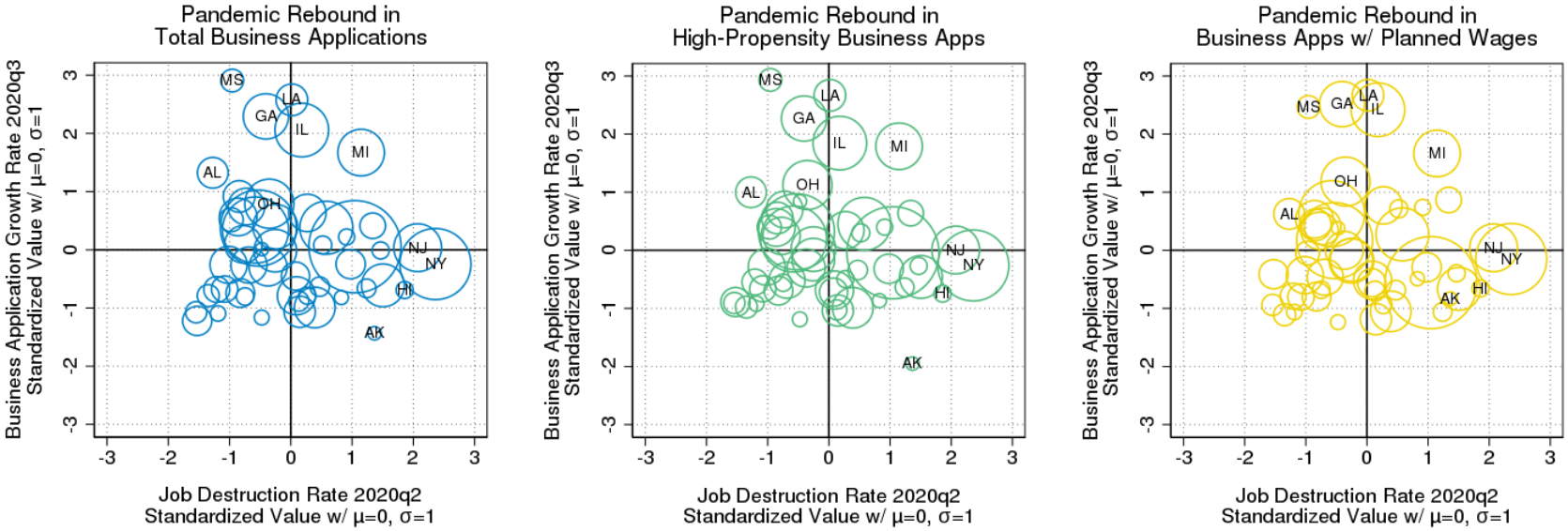
Note: This graph shows the actual difference between the MSA exit rate and the average exit rate across all MSAs for the 15 MSAs with the lowest exit rates. It also shows two counterfactual differences: one calculated using MSA-average sector-specific exit rates and one calculated using MSA-average share of firms in each sector. A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 20: Count of exiting firms by quarter and number of prior quarters active: 2007-2020



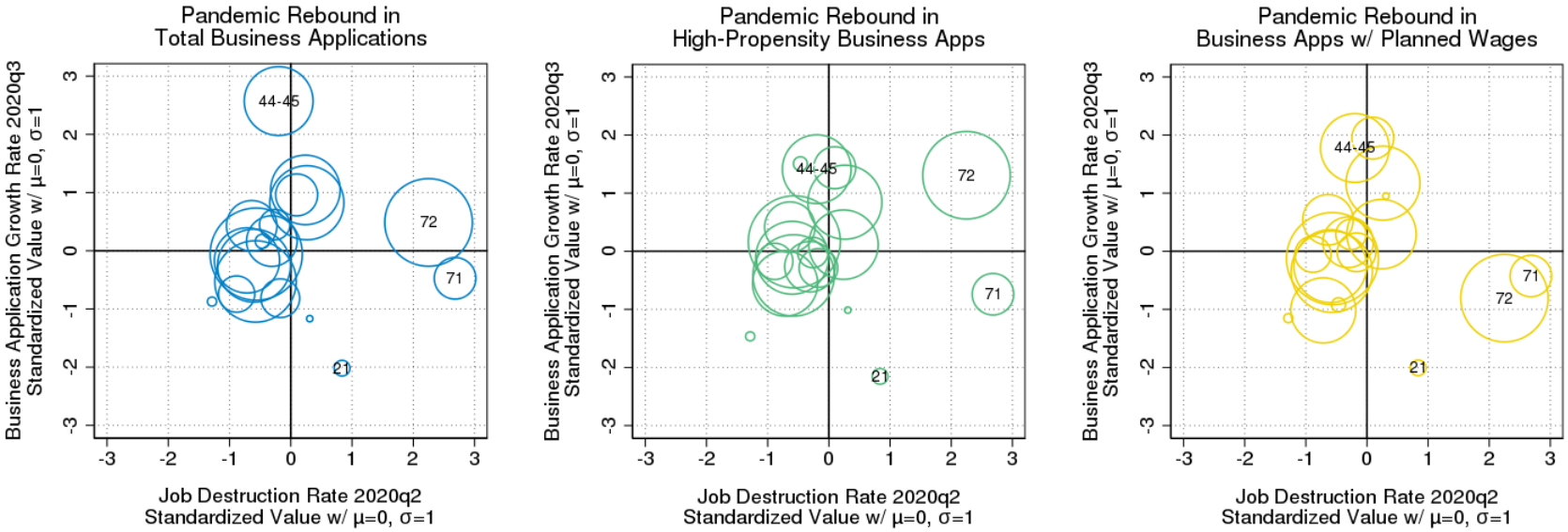
Note: A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. Number of prior quarters active counts how many previous quarters a firm had positive payroll and employment, prior to exiting. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 21: Job Destruction Rates and New Business Application Activity across States



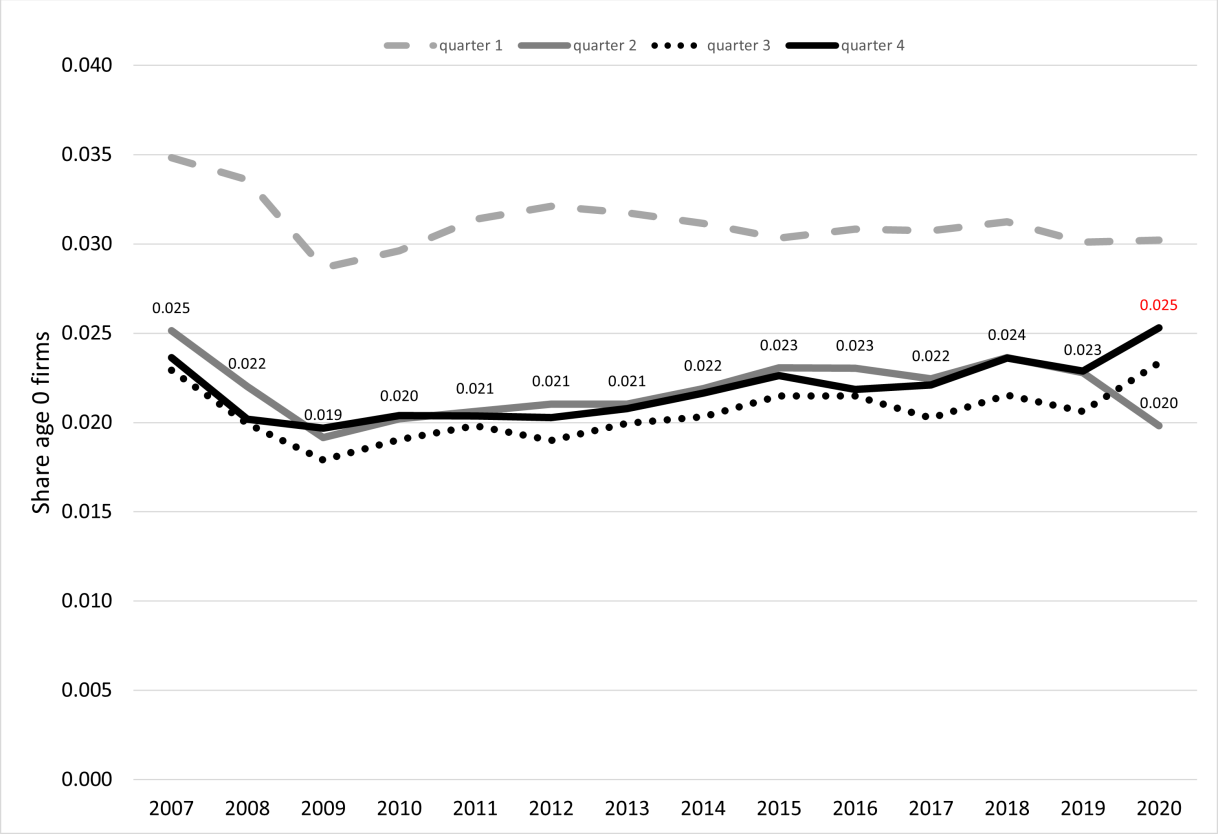
Note: This figure presents three scatter plots comparing the job destruction rate in 2020q2 and the subsequent business applications growth rate in 2020q3. All growth rates will be standardized with mean=0 and sd=1. Each bubble is weighted by denom. High-propensity business applications are classified as "likely employers." These applications have characteristics that make them more likely to transition into an employer business. Business applications with planned wages are a subset of the high-propensity type. The average job destruction rate across states in 2020q2 measured 23.0 percent and the average business application growth rate in the following quarter measured 58.6 percent. This translates into a job destruction level of 241,796 and a business application level of 28,940. This implies a business-application-per-job-lost ratio of roughly 12 percent.

Figure 22: Job Destruction Rates and New Business Application Activity across Sectors



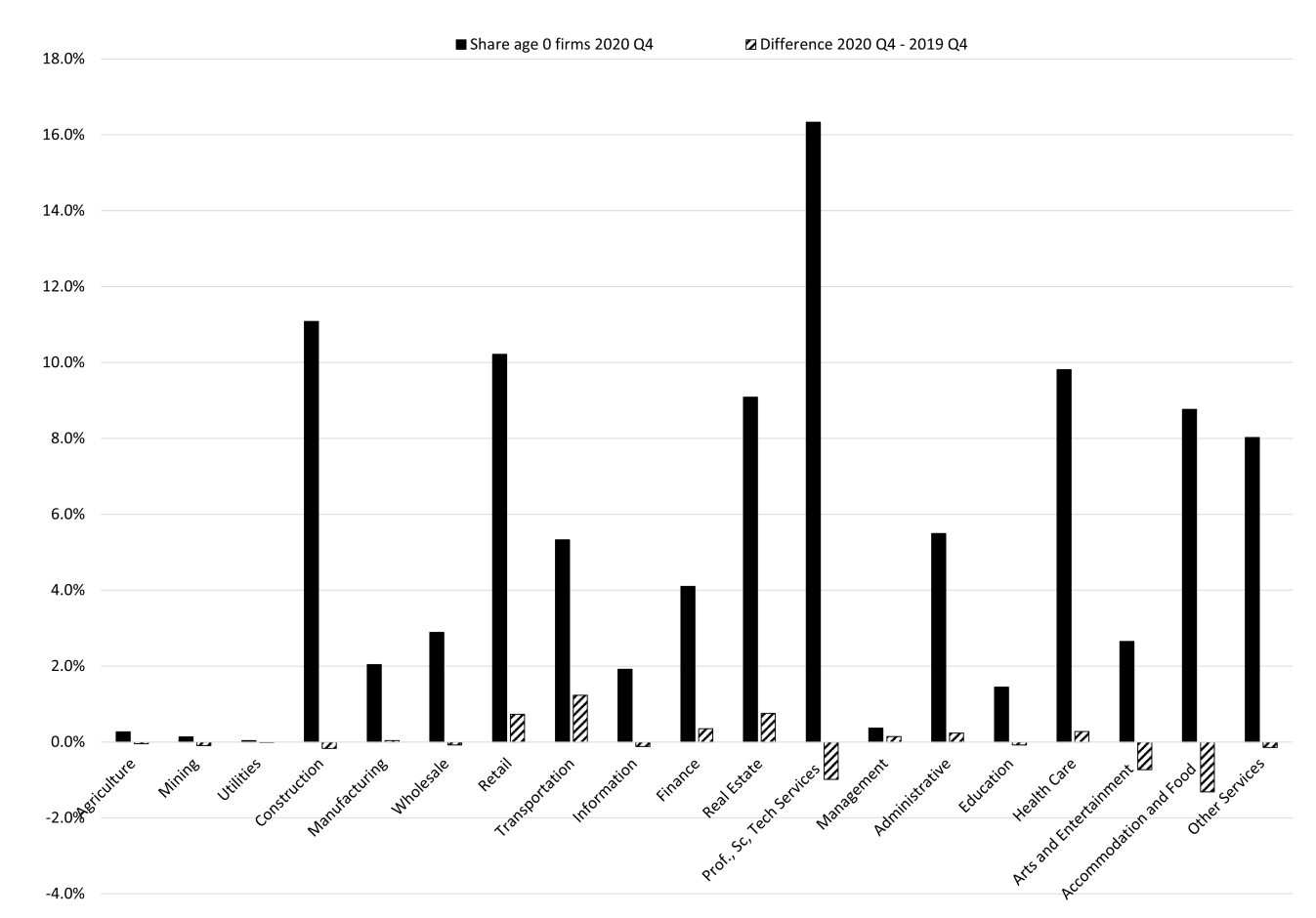
Note: This figure presents three scatter plots comparing the job destruction rate in 2020q2 and the subsequent business applications growth rate in 2020q3. All growth rates will be standardized with mean=0 and sd=1. Highlighted NAICS sectors are defined as follows: Retail Trade (44-45), Arts, Entertainment, and Recreation (71), and Accommodation and Food Services (72). Each bubble is weighted by denom. High-propensity business applications are classified as "likely employers." These applications have characteristics that make them more likely to transition into an employer business. Business applications with planned wages are a subset of the high-propensity type. The average job destruction rate across sectors in 2020q2 measured 23.0 percent and the average business application growth rate in the following quarter measured 55.6 percent. This translates into a job destruction level of 648,204 and a business application level of 76,236. This implies a business-application-per-job-lost ratio of roughly 11.8 percent.

Figure 23: Share of age 0 firms, by quarter: 2007-2020



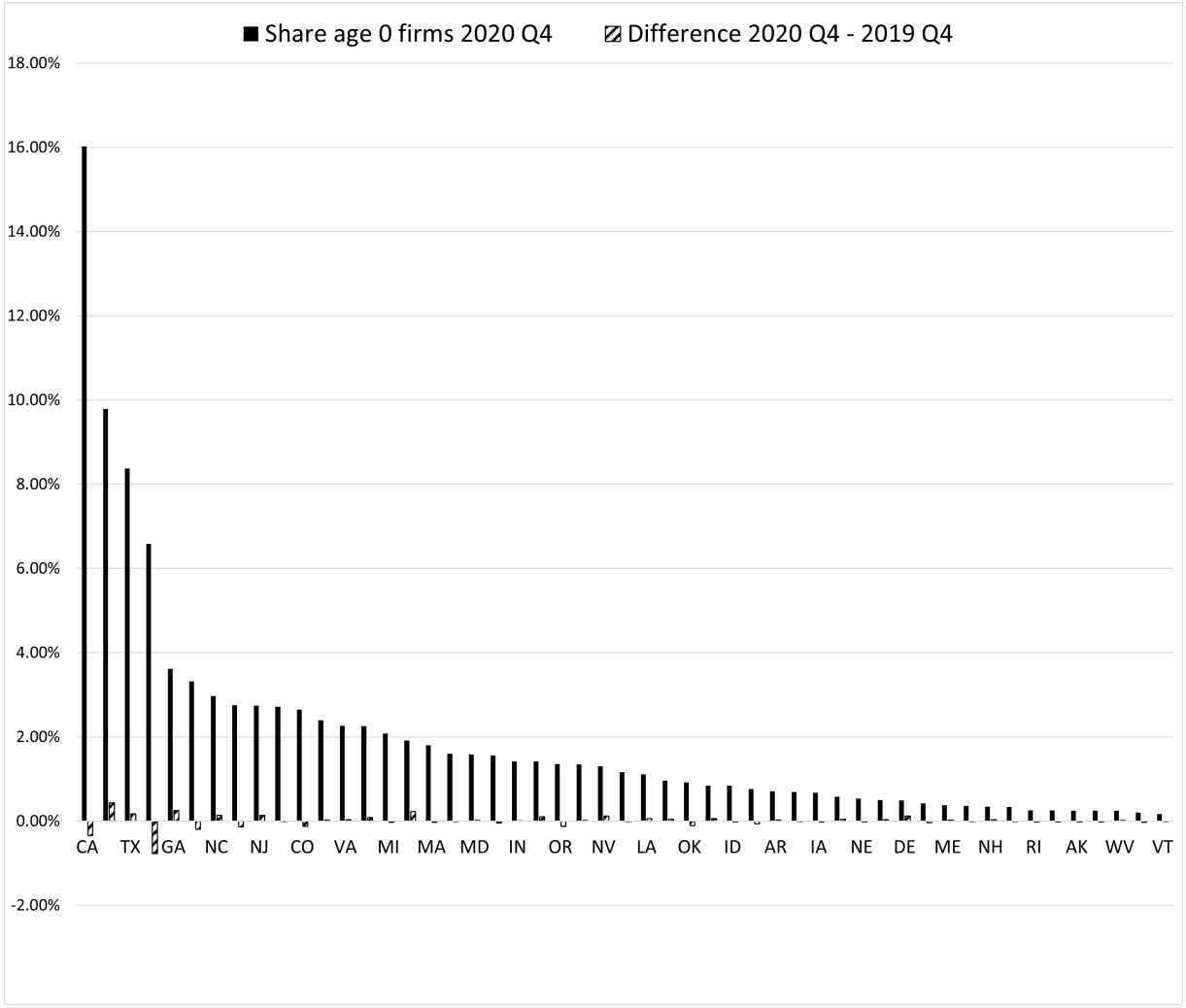
Note: A firm that reports positive payroll and employment for the first time ever in a given year and quarter is labeled as age 0, meaning it is a new firm or start-up. The share of age 0 firms is calculated as the ratio of total new firms to total firms of all ages operating in a given year, quarter. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 24: Change in share of age 0 firms in each NAICS sector



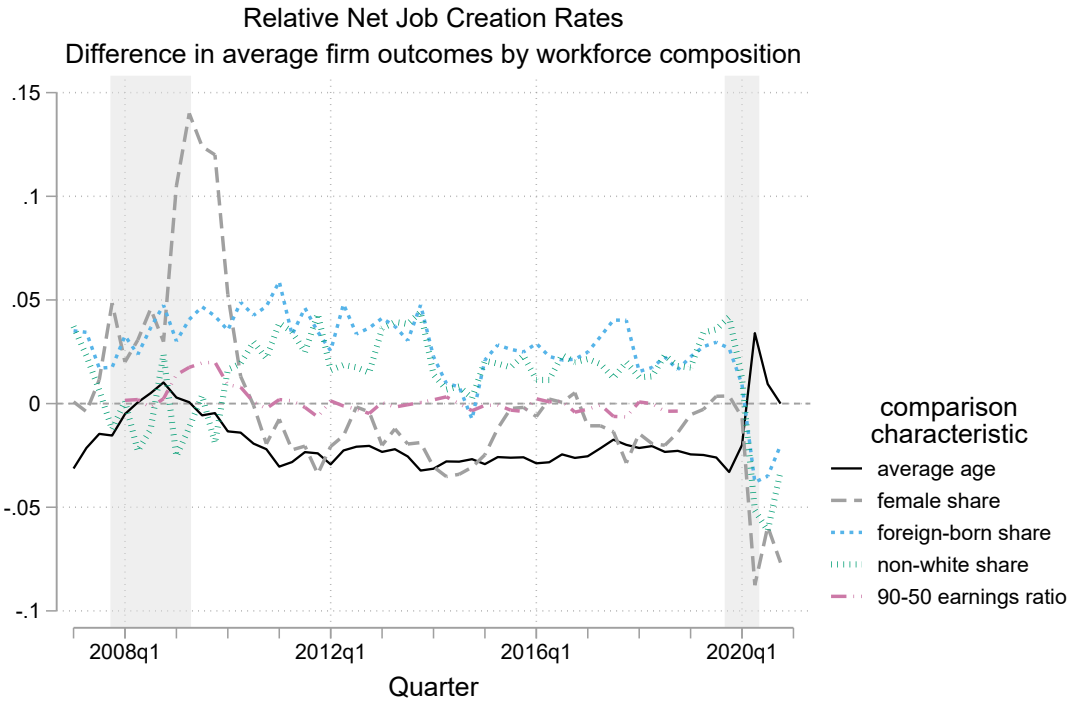
Note: A firm that reports positive payroll and employment for the first time ever in a given year and quarter is labeled as age 0, meaning it is a new firm or start-up. The share of age 0 firms is calculated as the ratio of total new firms to total firms of all ages operating in a given year, quarter. Industry sector is based on 2017 NAICS codes from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 25: Change in share of age 0 firms in each State



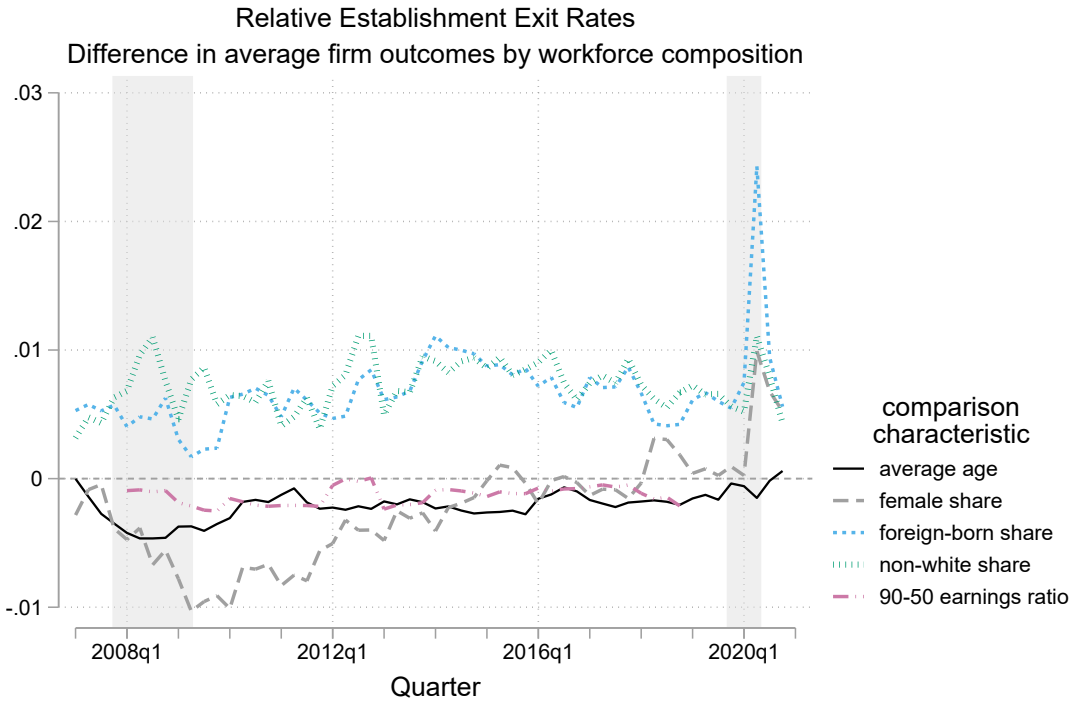
Note: A firm that reports positive payroll and employment for the first time ever in a given year and quarter is labeled as age 0, meaning it is a new firm or start-up. The share of age 0 firms is calculated as the ratio of total new firms to total firms of all ages operating in a given year, quarter. State is based on geocodes assigned to firms in the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release

Figure 26: Quarterly relative net job creation rates by firm workforce composition



Note: This figure plots quarterly estimates of the average difference in firm net job creation rates by each firm workforce characteristic. The estimation equation is Equation 4, and the interpretation of the estimates is the differential exposure of each type of worker to firm-level business dynamics. The sample is a random 10% sample of all firms in the BDS-SU universe for all workforce characteristics besides earnings inequality, and the subset of firms with at least 10 continuously employed workers and median annual stayer earnings of at least $5000 in 2012 dollars for the earnings inequality estimates. Average workforce age is in decades. The firm workforce female share, foreign-born share, and non-white (or Hispanic) share all range from 0 to 1. Earnings inequality is the ratio between the 90th percentile and the 50th percentile in annual earnings among continuously employed workers at each establishment. Regressions are weighted by average quarterly employment in the current and previous year (*bds_denom*).

Figure 27: Quarterly relative establishment exit rates by firm workforce composition



Note: This figure plots quarterly estimates of the average difference in establishment exit rates by each firm workforce characteristic. The estimation equation is Equation 4, and the interpretation of the estimates is the differential exposure of each type of worker to firm-level business dynamics. The sample is a random 10% sample of all firms in the BDS-SU universe for all workforce characteristics besides earnings inequality, and the subset of firms with at least 10 continuously employed workers and median annual stayer earnings of at least $5000 in 2012 dollars for the earnings inequality estimates. Average workforce age is in decades. The firm workforce female share, foreign-born share, and non-white (or Hispanic) share all range from 0 to 1. Earnings inequality is the ratio between the 90th percentile and the 50th percentile in annual earnings among continuously employed workers at each establishment. Regressions are weighted by average quarterly employment in the current and previous year (*bds_denom*).

Appendix C Tables

Table 1: Economy-wide Statistics for BDS-SU and BDS: 2020Q1

	(1)	(2)	(3)
Level stats	**BDS-SU**	**BDS**	**Ratio**
emp	52,662,126	132,833,423	0.40
estabs	5,148,756	7,162,267	0.72
estabs_entry	567,570	669,196	0.85
estabs_exit	569,610	675,130	0.84
job_creation	7,583,318	15,766,762	0.48
job_creation_births	2,735,923	5,223,039	0.52
job_creation_continuers	4,847,395	10,543,723	0.46
job_destruction	7,070,106	14,975,528	0.47
job_destruction_continuers	4,261,320	10,101,896	0.42
job_destruction_deaths	2,808,786	4,873,632	0.58
net_job_creation	513,212	791,234	0.65
Rate stats	**BDS-SU**	**BDS**	**Difference**
estabs_entry_rate	11.02%	9.34%	1.68%
estabs_exit_rate	11.06%	9.42%	1.64%
job_creation_rate	14.47%	11.91%	2.56%
job_creation_rate_births	5.22%	3.94%	1.28%
job_destruction_rate	13.49%	11.31%	2.18%
job_destruction_rate_deaths	5.36%	3.68%	1.68%
net_job_creation_rate	0.98%	0.60%	0.38%

Notes: This table reports select statistics for 2020q1 from the economy-wide BDS-SU table in Column 1, as well as the corresponding value from the core BDS in Column 2. Column 3 reports either the ratio of the BDS-SU value to the BDS value (for level statistics in the top panel), or the difference between the two (for rate statistics in the bottom panel).

Table 2: Estab. and Emp. Shares for BDS-SU and Core BDS: 2020Q1

	(1)	(2)	(3)	(4)
Category	**Estab share BDS-SU**	**Estab share BDS**	**Emp share BDS-SU**	**Emp share BDS**
Age 0	0.08	0.08	0.04	0.04
Age 1-5	0.24	0.26	0.18	0.17
Age 6-10	0.17	0.16	0.16	0.14
Age 11+	0.51	0.50	0.62	0.66
Size 1-9	0.78	0.71	0.24	0.13
Size 10-99	0.21	0.27	0.50	0.39
Size 100-499	0.01	0.02	0.17	0.25
Size 500+	0.00	0.00	0.09	0.22
Metro	0.95	0.94	0.96	0.96
Nonmetro	0.05	0.05	0.04	0.03
State-Wide	0.00	0.00	0.00	0.01
Unclassified	0.00	0.00	0.00	0.00
Agriculture (11)	0.00	0.00	0.00	0.00
Mining (21)	0.00	0.00	0.00	0.00
Utilities (22)	0.00	0.00	0.00	0.00
Construction (23)	0.12	0.09	0.10	0.05
Manufacturing (31-33)	0.04	0.04	0.07	0.09
Wholesale Trade (42)	0.04	0.05	0.04	0.05
Retail Trade (44-45)	0.10	0.13	0.08	0.12
Transportation (48-49)	0.03	0.03	0.03	0.04
Information (51)	0.01	0.02	0.01	0.03
Finance (52)	0.04	0.06	0.02	0.05
Real Estate (53)	0.05	0.05	0.02	0.02
Professional (54)	0.13	0.11	0.09	0.07
Management (55)	0.00	0.01	0.00	0.03
Administrative (56)	0.06	0.05	0.09	0.09
Education (61)	0.02	0.01	0.05	0.03
Health Care (62)	0.11	0.12	0.14	0.16
Arts and Ent. (71)	0.02	0.02	0.03	0.02
Accom. and Food (72)	0.09	0.09	0.13	0.11
Other Services (81)	0.12	0.10	0.08	0.04

Notes: This table reports the share of establishments and employment belonging to various categories based on establishment age, size, metropolitan classification, and 2-digit NAICS industry, using measures from 2020q1. Columns 1 and 2 report the share based on the number of establishments from the relevant BDS-SU tables and core BDS economy-wide tables respectively. Columns 3 and 4 similarly show the corresponding shares based on employment. Each column sums to 1.00 within each sub-panel.

Table 3: Quarterly Pay Statistics in BDS-SU: 2020Q1

	(1)	(2)
Statistic	**Nominal**	**Real**
pay	572,864,650	505,412,322
pay_creation	91,162,798	76,219,208
pay_creation_births	21,675,302	19,123,129
pay_creation_continuers	69,487,496	57,096,079
pay_destruction	62,184,957	58,536,264
pay_destruction_continuers	23,905,318	21,437,055
pay_destruction_deaths	38,279,639	21,437,055
net_pay_creation	28,977,841	17,682,944
pay_emp	10.878	9.697
pay_emp_continuers	11.040	9.740
pay_emp_births	7.922	6.990
pay_emp_deaths	8.511	7.632
pay_creation_rate	16.33%	15.35%
pay_creation_rate_births	3.88%	3.85%
pay_destruction_rate	11.14%	11.79%
pay_destruction_rate_deaths	4.28%	4.32%
net_pay_creation_rate	5.19%	3.56%

Notes: This table reports payroll statistics from the economy-wide BDS-SU tables for 2020q1. Column 1 expresses the values in nominal terms; Column 2 expresses pay in real, inflation-adjusted terms, calculated using the Bureau of Labor Statistics' GDP-deflator based on 2012 dollars. All dollar values are in reported in thousands.

Table 4: Metropolitan Statistical Areas Exit Rates: Highest 15 cities

MSAs ranked by 2020 Q2 establishment exit rate: top 15 National establishment exit rate=15.105			
MSA	exit rate	diff. from 2019 Q2	diff. from national 2020 Q2 exit rate
New York-Newark-Jersey City, NY-NJ-PA	24.23	13.83	9.13
Atlantic City-Hammonton, NJ	22.01	12.77	6.90
Kahului-Wailuku-Lahaina, HI	21.13	12.52	6.03
Detroit-Warren-Dearborn, MI	20.96	11.23	5.86
East Stroudsburg, PA	19.17	10.74	4.07
Trenton, NJ	19.00	8.80	3.90
Kingston, NY	18.90	9.59	3.79
San Jose-Sunnyvale-Santa Clara, CA	18.82	7.60	3.72
Vineland-Bridgeton, NJ	18.82	10.19	3.72
San Francisco-Oakland-Hayward, CA	18.44	7.84	3.34
Ocean City, NJ	18.34	9.91	3.23
Philadelphia-Camden-Wilmington, PA-NJ-DE-MD	18.32	9.08	3.22
Los Angeles-Long Beach-Anaheim, CA	18.28	7.33	3.17
Allentown-Bethlehem-Easton, PA-NJ	18.27	9.62	3.17
Flint, MI	18.00	8.61	2.89

Note: This table reports the 15 MSAs with the highest firm exit rates in 2020 quarter 2. A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release.

Table 5: Metropolitan Statistical Areas Exit Rates: Lowest 15 cities

MSAs ranked by 2020 Q2 establishment exit rate: bottom 15 National establishment exit rate=15.105			
MSA	exit rate	diff. from 2019 Q2	diff. from national 2020 Q2 exit rate
Amarillo, TX	10.11	1.67	-5.00
Hickory-Lenoir-Morganton, NC	10.10	1.60	-5.01
Ames, IA	10.01	0.97	-5.10
Grand Forks, ND-MN	10.00	1.63	-5.11
Sebring, FL	9.98	0.35	-5.13
Dalton, GA	9.93	1.41	-5.17
Appleton, WI	9.90	1.82	-5.21
Valdosta, GA	9.83	-0.19	-5.28
Elkhart-Goshen, IN	9.66	2.29	-5.45
St. Cloud, MN	9.50	2.59	-5.60
Fond du Lac, WI	9.46	2.83	-5.64
Mankato-North Mankato, MN	9.35	1.71	-5.76
Sioux Falls, SD	9.20	0.76	-5.91
Wausau, WI	9.01	2.65	-6.09
La Crosse-Onalaska, WI-MN	8.85	1.20	-6.26

Note: This table reports the 15 MSAs with the lowest firm exit rates in 2020 quarter 2. A firm exits when it has no payroll or employment in a given quarter of a year but had positive payroll and employment in the same quarter of the prior year. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release.

Table 6: Decomposition of MSA exit rates

Effect of MSA sector exit rates versus MSA sector shares on overall MSA exit rate		
MSA exit rate above average rate across all MSAs		
	sector shares push MSA rate	
sector rates push MSA rate	down	up
down	0	14
up	15	35
MSA exit rate below average rate across all MSAs		
	sector shares push MSA rate	
sector rates push MSA rate	down	up
down	140	175
up	2	0

Note: This table reports whether an MSA was above or below the national average exit rate; whether exposure to certain industries (local share in each sector) pushed the MSA exit rate below or above the national average; whether exit rates in certain industries (local exit rates) pushed the MSA exit rate below or above the national average. The exit rate is the number of exiting firms divided by the average of total firms operating in the current and the prior year. Firms are assigned to vintage 2012 MSA definitions based on geography information from the Census Bureau Business Register. Source: Business Dynamics Statistics Single Unit Firms, 2022 release.

Table 7: Relative net job creation rates by firm workforce composition

Dep. Variable	Net job creation rate			
	(1)	(2)	(3)	(4)
Average age	-0.0199***	-0.0232***	-0.0228***	-0.0122***
	(0.00173)	(0.00166)	(0.00173)	(0.00243)
Female share	-0.00144	-0.0292***	-0.0289***	-0.00428
	(0.00537)	(0.00402)	(0.00389)	(0.00597)
Foreign-born share	0.0408***	0.0417***	0.0476***	0.0410***
	(0.00476)	(0.00383)	(0.00397)	(0.0113)
Non-white share	-0.0114*	-0.0168***	-0.0229***	-0.0435***
	(0.00471)	(0.00440)	(0.00467)	(0.00696)
90-50 earnings ratio				-0.00118
				(0.00117)
Rounded Obs.	31180000	31180000	31180000	2927000
Adjusted R-sq.	0.00573	0.00848	0.00903	0.0136
Quarter FE	Yes	Yes	Yes	Yes
NAICS FE		Yes	Yes	Yes
State by MSA FE			Yes	Yes

Note: This table reports estimates of the average difference in firm net job creation rates by firms' workforce characteristics. The estimation equation is Equation 5, and the interpretation of the estimates is the differential exposure of each type of worker to firm-level business dynamics. The sample in the first three columns is a random 10% sample of all firms in the BDS-SU universe, and the sample in the fourth column is the subset of firms with at least 10 continuously employed workers and median annual stayer earnings of at least $5000 in 2012 dollars. Average workforce age is in decades. The firm workforce female share, foreign-born share, and non-white (or Hispanic) share all range from 0 to 1. Earnings inequality is the ratio between the 90th percentile and the 50th percentile in annual earnings among continuously employed workers at each establishment. Regressions are weighted by average quarterly employment in the current and previous year (*bds_denom*), and standard errors are clustered by industry. Stars indicate the level of statistical significance: * 5%, ** 1%, and *** 0.1%.

Table 8: Relative establishment exit rates by firm workforce composition

Dep. Variable	Establishment exit rate			
	(1)	(2)	(3)	(4)
Average age	-0.00197*	0.00118***	0.00115***	-0.00222***
	(0.000876)	(0.000290)	(0.000298)	(0.000474)
Female share	-0.00263	-0.00198	-0.00212	-0.00315
	(0.00233)	(0.00118)	(0.00117)	(0.00184)
Foreign-born share	0.00254	-0.00290*	-0.00423***	-0.0126***
	(0.00139)	(0.00126)	(0.00128)	(0.00212)
Non-white share	0.00561***	0.00731***	0.00639***	0.0108***
	(0.00120)	(0.000715)	(0.000885)	(0.00205)
90-50 earnings ratio				-0.00134***
				(0.000188)
Rounded Obs.	31180000	31180000	31180000	2927000
Adjusted R-sq.	0.000652	0.00620	0.00700	0.00862
Quarter FE	Yes	Yes	Yes	Yes
NAICS FE		Yes	Yes	Yes
State by MSA FE			Yes	Yes

Note: This table reports estimates of the average difference in establishment exit rates by firms' workforce characteristics. The estimation equation is Equation 5, and the interpretation of the estimates is the differential exposure of each type of worker to firm-level business dynamics. The sample in the first three columns is a random 10% sample of all firms in the BDS-SU universe, and the sample in the fourth column is the subset of firms with at least 10 continuously employed workers and median annual stayer earnings of at least $5000 in 2012 dollars. Average workforce age is in decades. The firm workforce female share, foreign-born share, and non-white (or Hispanic) share all range from 0 to 1. Earnings inequality is the ratio between the 90th percentile and the 50th percentile in annual earnings among continuously employed workers at each establishment. Regressions are weighted by average quarterly employment in the current and previous year (*bds_denom*), and standard errors are clustered by industry. Stars indicate the level of statistical significance: * 5%, ** 1%, and *** 0.1%.

Table 9: Relative net job creation rates (excluding exit) by firm workforce composition

Dep. Variable	Net job creation rate (excluding exit)			
	(1)	(2)	(3)	(4)
Average age	-0.0238***	-0.0209***	-0.0205***	-0.0167***
	(0.00162)	(0.00159)	(0.00168)	(0.00202)
Female share	-0.00671	-0.0331***	-0.0331***	-0.0106*
	(0.00628)	(0.00518)	(0.00490)	(0.00476)
Foreign-born share	0.0458***	0.0359***	0.0392***	0.0158
	(0.00506)	(0.00479)	(0.00439)	(0.0101)
Non-white share	-0.000210	-0.00220	-0.0101*	-0.0219**
	(0.00540)	(0.00481)	(0.00480)	(0.00690)
90-50 earnings ratio				-0.00387***
				(0.00107)
Rounded Obs.	31180000	31180000	31180000	2927000
Adjusted R-sq.	0.00739	0.0131	0.0142	0.0186
Quarter FE	Yes	Yes	Yes	Yes
NAICS FE		Yes	Yes	Yes
State by MSA FE			Yes	Yes

Note: This table reports estimates of the average difference in firm net job creation rates (excluding exit) by firms' workforce characteristics. Exiting establishments are included in the regression but have a net job creation rate of zero, so the coefficients are directly comparable to the coefficients in Table 7. The estimation equation is Equation 5, and the interpretation of the estimates is the differential exposure of each type of worker to firm-level business dynamics. The sample in the first three columns is a random 10% sample of all firms in the BDS-SU universe, and the sample in the fourth column is the subset of firms with at least 10 continuously employed workers and median annual stayer earnings of at least $5000 in 2012 dollars. Average workforce age is in decades. The firm workforce female share, foreign-born share, and non-white (or Hispanic) share all range from 0 to 1. Earnings inequality is the ratio between the 90th percentile and the 50th percentile in annual earnings among continuously employed workers at each establishment. Regressions are weighted by average quarterly employment in the current and previous year (*bds_denom*), and standard errors are clustered by industry. Stars indicate the level of statistical significance: * 5%, ** 1%, and *** 0.1%.